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Predicting the probabilistic dispersion of suborbital unguided rockets

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List of Acronyms

CAA Civil Aviation Authority.
CATO Catastrophe At Take Off.
CEP Circular Error Probable.

DoF Degree of Freedom.

GFS Global Forecast System.

ISA International Standard Atmosphere.
ISO International Organization for Standardization.

LVSEF Launch Vehicle and Space Early Failure.

NAR National Association of Rocketry.
NASA National Aeronautics and Space Administration.
NOTAM Notice to Air Missions.

PRA Probabilistic Risk Assessment.

UKRA United Kingdom Rocketry Association.

1 Abstract

The rapid growth of student rocketry in the UK has increased launch ambitions and vehicle performance, but safety and dispersion analysis methods have not kept pace. As a result, uncertainty in the landing dispersion of unguided suborbital rockets remains a key limitation in UK launch site planning and operational constraints. The primary aim is to investigate the key factors influencing dispersion in suborbital unguided rockets through dispersion modelling and sensitivity analysis, with a focus on improving safety-case fidelity for launch operations and range constraints.

A stochastic Monte Carlo model was developed based on an existing 6DoF rocket trajectory simulator and incorporates modelling of flight-failure events to estimate landing dispersion. This model was then experimentally validated by comparing the dispersion analysis of a rocket to its actual landing location. The framework was then applied to both a high-altitude and a low-altitude case study. A sensitivity analysis was then conducted to determine the relative contribution of key uncertainty sources, including atmospheric conditions, launch configuration, and vehicle parameters, to overall dispersion behaviour.

The model was successfully validated against experimental flight data, demonstrating reasonable agreement between predicted and observed dispersion. Results show that dispersion is most sensitive to wind uncertainty, followed by launcher uncertainty for high-altitude launches. For low-altitude launches, launch-related uncertainties become the dominant factor, followed by wind uncertainties. Additionally, high-energy ballistic failure modes produce the most extreme landing cases, often occurring outside the 98.9% confidence areas, representing a significant hazard to people and property. These results highlight the importance of improving failure-mitigation and impact-reduction strategies alongside reducing dispersion uncertainty, particularly for larger rocket systems. Additionally, it's been demonstrated that wind uncertainty can be estimated from flight data and used to predict the landing location of the same rocket.

This work enhances understanding of how input uncertainty, including failures, affects rocket dispersion and provides a foundation for future research in the field of rocket dispersion analysis and probabilistic casualty models. Likewise, these methods can be applied to rockets and launches to support evidence-based decision-making.

Lay Summary

Student rocketry in the UK is growing and becoming more advanced, but predicting exactly where a rocket will land is difficult. This is because it's very difficult to accurately estimate all the factors that affect the landing location, such as wind, as well as small variations in how rockets are launched and how they behave during flight. Because of this, rocketry teams must use large safety margins to ensure flights are safe, which limits where and how high rockets can fly.

This project predicts where rockets are likely to land by building a computer model that simulates a very large number of rocket flights with subtly different inputs to account for factor uncertainty. The model is tested against a real rocket flight to verify that it works, and then different inputs are varied to estimate which has the greatest impact on the size of the landing area. This identified wind and the rocket's initial angle as the factors with the greatest impact on the landing area. A technique for estimating where the rocket will land based on wind information from the rocket launch itself is shown as a method to reduce the estimated landing area for successive flights.

The results help improve understanding of where rockets are likely to land and show factors that matter most for safety. This helps show that a rocket launch will be safer and could enable flights in new places and to higher altitudes.

2 Introduction

In recent years, there has been a considerable increase in the number and ambition of student rocketry teams within the UK [1]. This has led to increased development and understanding of most aspects of rocket design, including propulsion, aerostructures, control, and avionics, through the UKSEDS MachX, Race2Space, Launch4Change, and European Rocketry Challenge (EuRoC) competitions. However, far fewer development and research efforts have been directed towards the operation of launch sites, safety cases for these rockets, and a better understanding of dispersion areas, as these are typically handled by the teams organising the competitions or launch events rather than the launch teams themselves. This has led to a situation in which rocketry teams continue to push the launch boundaries as they innovate, flying higher and faster without a deeper understanding of dispersion patterns, resulting in conservative limits around launch sites due to uncertainty about the probable landing location and, consequently, the underlying safety case.

The motivation for this project is a desire to better understand where rockets will land during a launch at Llanbedr Airfield. Principally using existing rocket simulation programs in combination with weather reports, we were unable to accurately predict the landing locations of our rockets. This made it difficult to determine the operational limits and design decisions to ensure the rockets landed within the site bounds. During the launches, this was attributed to variable and unpredictable winds and subtle inaccuracies in the rocket's modelling. The uncertainty in the probable landing locations led to conservative limits to maintain the safety case for the launches.

Throughout the UK, launch altitudes and opportunities are typically limited by available landing areas, both at large organised events and smaller outreach activities [2]. The probable landing locations of unguided rockets are difficult to determine precisely due to uncertainties in atmospheric conditions and vehicle dynamics [3]. This project aims to investigate the factors influencing dispersion in suborbital unguided rockets through trajectory and sensitivity analysis. This leads us to define the following objectives:

- Develop a simulation model to predict the flight trajectories and impact dispersion of suborbital unguided rockets.
- Quantify the influence of local atmospheric conditions, including wind, on rocket trajectory dispersion.
- Implement stochastic modelling techniques to represent uncertainties arising from flight-event variability and system perturbations.
- Validate and evaluate the predictive accuracy of the simulation model using experimental or recorded flight-test data.
- Design and implement an effective visualisation framework for communicating trajectory uncertainty and dispersion characteristics.

3 Literature Review

To investigate the project's aims, this section presents a thorough literature review of the background theory, relevant datasets and existing approaches to rocket trajectory and dispersion analysis. This section focuses on the current understanding of suborbital rocket flight dynamics, launch range safety, atmospheric modelling, deterministic trajectory simulation, stochastic uncertainty modelling, validation approaches, and dispersion visualisation techniques. Through critical analysis of existing methods and tools, the review identifies limitations in current approaches to probabilistic trajectory prediction and establishes the motivation for the Monte Carlo-based methodology developed within this project.

3.1 Rocket Flight and Recovery Dynamics

The understanding of rocket flight and recovery dynamics is critical in understanding how rockets will perform and ultimately predicting where they will land. This report focuses on sub-orbital unguided rockets with a single engine, which return to the ground without reaching orbit, have no active steering or control, and use a single motor, as these are the most common types of rockets launched in the UK.

3.1.1 Rocket Fundamentals

Rockets generate thrust by expelling high-speed mass from an engine, often described as a motor, through combustion, to propel the vehicle forward. Unguided rockets are a subset of rockets that rely entirely on passive mechanisms for attitude control. As seen in Figure 3.1, these rockets typically comprise a nose cone designed to minimise drag, a cylindrical body, fins for stability, a motor for thrust, and recovery systems to safely recover the rocket and any onboard payloads.

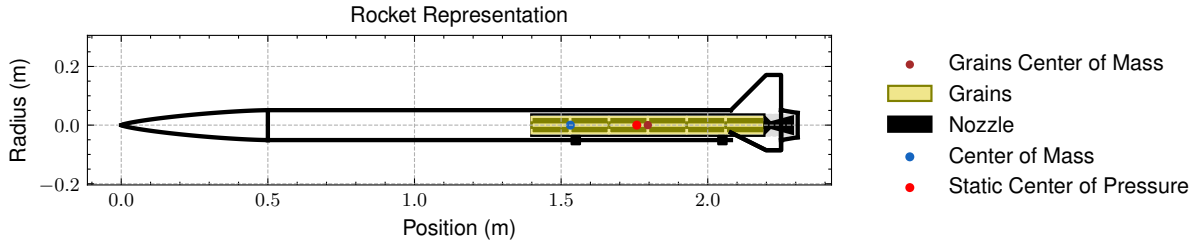


Figure 3.1: Unguided rocket with aerodynamic structure and rocket motor.

These rockets use the principle of positive static stability to provide a restoring force that keeps them flying straight when forces such as wind gusts act on them. This works as the fins produce lift from the airflow over the rocket, providing a restoring moment to counteract the disturbances. The strength of this restoring depends on a

combination of fin and rocket geometry, rocket mass and mass distribution, and air speed and density.

The lift force, L, is given by the following equation, where ρ is density, V is airspeed, S is fin reference area, and C_L is the coefficient of lift for the fin.

$$L = \frac{1}{2}\rho V^2 S C_L \tag{3.1}$$

Static margin is a dimensionless quantity that is used to assess the static stability of rockets, where positive values of static margin are positively stable, and values between 1.5 and 3 [4] are typical for subsonic rockets to provide sufficient restoring force whilst not being unstable, and therefore uncontrollable or over-stable. The static margin depends on the location of the centre of gravity, x_{CG}, the location of the centre of pressure, x_{CP}, and the rocket diameter, D. The rocket mass and geometry (including fins) are chosen to meet this stability criterion using rocket modelling software.

$$\text{Static Margin} = \frac{x_{CG} - x_{CP}}{D} \tag{3.2}$$

Over-stable rockets exhibit a phenomenon called weathercocking, where they turn into the wind rather than flying straight up. This is due to the restoring moment, working towards the effective flow direction, comprising the resultant vector of the wind velocity, w and the rocket's relative velocity, V, described as below:

$$\alpha = 90 - \arctan(\frac{V}{w}) \tag{3.3}$$

Here, α is the angle of attack (angle between the rockets' relative flow and the wind velocity). The relationships between airspeed, wind speed, and lift can be determined for the symmetric aerofoils that make up the fins.

$$C_L = a\alpha \tag{3.4}$$

This shows that the lift generated by the fins provide a restoring moment that rotates the rocket towards the effective flow direction, with a magnitude proportional to $\arctan(V/w)$ and therefore very sensitive to high wind speeds and low rocket airspeeds. This effect can significantly alter a flight's profile, affecting the landing distance and the altitude reached.

Rockets are recovered using high-drag devices such as parachutes or streamers. These slow the rockets down so they land slowly enough to avoid impact damage

and injury to people or property. The devices are usually deployed with a fixed delay after motor burnout (motor ejection) or by a dedicated flight computer at a designated point in the flight. Larger parachutes lead to a slower descent rate and an increase in the distance the rocket drifts. Whereas, small parachutes drift less but descent faster, leading to a direct trade-off that can constrain launches within safety constraints. One approach to address this is to use 2 parachutes: a low-drag drogue parachute deployed at apogee to descend quickly but controllably, and then a second, larger main parachute to land at a safe speed whilst reducing the drift distance, as seen in Figure 3.2.

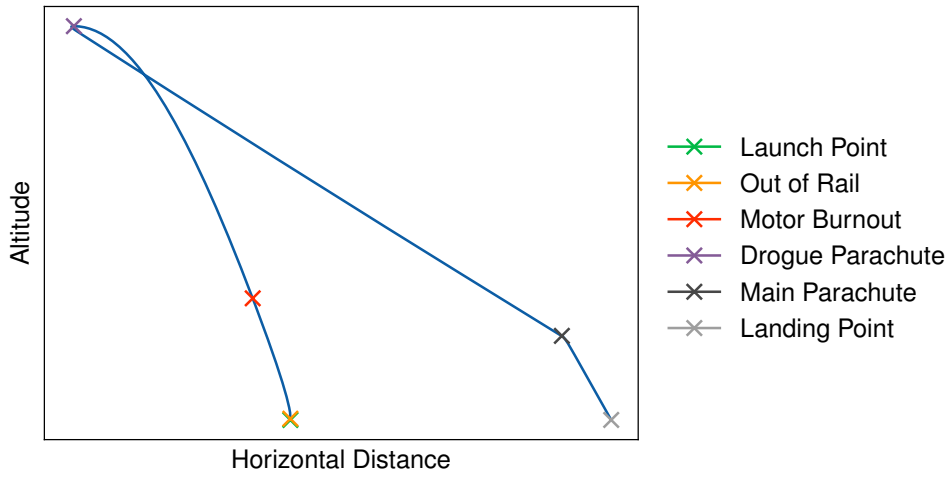


Figure 3.2: Flight trajectory of a typical unguided rocket flight with dual deployment recovery.

Each rocket flight comprises the following flight stages, as illustrated in Figure 3.2:

- **Ignition:** Rocket motors ignite on a launch rail (or tower) and accelerate to leave the rail (Out of Rail) with enough speed to be stable.
- **Powered Flight:** As the motor powers the rocket, it accelerates upwards.
- **Coast Phase:** The motor stops burning (burnout) and the rocket coast towards apogee.
- **Ejection:** The recovery system is deployed. This usually occurs around apogee.
- **Descent:** The rocket descends in a controllable manner whilst drifting with the wind. A second recovery system can be used to further slow the descent.
- **Landing:** The rocket touches down on the ground.

3.1.2 Rocket Failures

Understanding rocket failure modes is essential for predicting system performance under off-nominal conditions, assessing overall dispersion patterns and failure probability. Failure modes and probabilities in rockets are normally assessed using a component-based approach, such as in the NASA Probability Risk Assessment (PRA) Procedure

Guide, in which the failure modes and probabilities of individual components are determined and then modelled to estimate the probability of the entire system [5]. This is a resource-intensive process that requires understanding the failure performance of individual components. As such, this is not well-suited to this project due to the lack of component-level performance data and the need to conduct a PRA for each rocket. Although it is the standard approach for determining failure modes and performance, a generalised approach is preferred.

Robert B. Cross introduces a new approach that aims to better predict the failure probability of new orbital launch vehicles from historical data, whilst correctly observing that PRAs typically under-predict new launch vehicle failure probability because they model a mature vehicle in which a significant number of early failures have been eliminated through testing [6]. This approach determines the failure probability by design element, enabling the failure probability of a new rocket to be predicted based on its design elements and complexity. This approach would be ideal for this project and would enable the failure probability to be determined based on the design complexity; however, it is not suitable, as no verification data is available, and the underlying Launch Vehicle and Space Early Failure (LVSEF) database could not be found. Additionally, developing a new database using Jonathan C. McDowell's rocket launch database was deemed beyond the scope of this project [7].

The Launching Safely in the 21st Century report, published by the National Association of Rocketry (NAR), is a comprehensive review of model rocket flights up to a total impulse of 40960 Ns, which represents most student and amateur rockets [8]. This report includes an analysis of 6169 flights from rocket launch records and uses this data to assess the failure probability by motor class, as seen in Table 3.1. It's worth noting that this data covers 1998 to 2005 and may not be representative of current rocket technology however it is the most recent data available. This failure dataset also categorises all observed failures into the following categories:

- **Unstable**: The rocket flies with at least part of the boost phase in a nose-down attitude.
- **Lawn dart**: The rocket descends in ballistic flight with the nose cone still on.
- **Separation**: The rocket descends in multiple parts, with at least one part not slowed by a recovery device.
- **Motor CATO**: The motor fails catastrophically at ignition or during boost.
- **Core sample**: The rocket descends in ballistic flight, but with the nose cone off, acting as a (not very effective) drag device.
- **Motor unrestrained**: The motor exits the rocket at ignition or during boost (thankfully, this was rare).
- **Shred**: The rocket comes apart during ascent, other than by design.

- **No chute**: The rocket descends without a recovery device deployed, but is not ballistic.

Table 3.1: Failure probability percentages for single motor rocket by rocket motor class and failure mode from the NAR Report

	MM	1/4A	1/2A	A	B	C	D	E	F	G	H
Lawn Dart	3.2	3.7	2.1	0.4	0.3	1.2	2.0	4.8	0.7	1.7	0.0
Core Sample	0.0	0.0	0.0	0.3	0.2	0.2	0.7	1.3	0.4	0.0	3.5
Separation	0.0	3.7	2.7	2.6	3.6	2.5	2.6	1.3	1.1	1.7	0.0
Shred	0.0	0.0	1.1	0.3	0.1	0.2	0.2	0.0	0.0	0.0	0.0
Motor CATO	0.0	0.0	0.0	0.7	0.0	0.4	0.8	0.4	0.7	2.1	1.8
No chute	3.2	0.0	0.5	1.7	1.1	1.1	1.9	0.9	0.7	1.7	1.8
Unstable	13.0	0.0	0.5	1.8	1.5	1.5	1.1	2.6	0.7	0.0	0.0

Although the NAR report focuses on rockets smaller than the subject of this project, these rockets are more similar in approach than other sources, as they are mostly unguided. Additionally, the probability given by the motor class enables the application of this approach to smaller rockets with different failure performance. As such, this approach is chosen, and the failure probabilities of I-class and larger motors are assumed to be the same as H, with the caveat that the probabilities are extrapolated from the best available data. Whilst the report provides a relative risk table that characterises the risk of a bad outcome for a given failure mode and motor class, this project aims to use a more robust approach in which risk is determined from impact energy.

3.2 Launch Site Constraints and Range Safety

Understanding range safety and launch site constraints is critical to identify how they constrain rocket design and operation, and for providing context for the risk element associated with dispersion analysis. Commercial vertical launch sites, such as Saxa-Vord, will be excluded from this review because they aren't accessible to amateur and student teams.

Non-commercial launch sites in the UK follow the rules and regulations of the United Kingdom Rocketry Association (UKRA) and the UKRA Safety Code [9]. Under UKRA regulations and oversight, event organisers and participants can purchase liability insurance and larger motors, enabling them to launch. The launch organisers seek permission from landowners to launch and recover rockets from their land and liaise with the Civil Aviation Authority (CAA) to request permission to use airspace and issue a Notice To Air Missions (NOTAM) to warn other airspace users of launch operations. This forms the legal and safety basis for a rocket launch.

UKRA defines the minimum launch range diameter as the maximum of half the expected altitude and the diameter in Table 3.2 that corresponds to the rocket's total impulse

[2]. The highest non-commercial launch site is the Scottish Aeronautics and Rocketry Association's site on Fairlie Moor, Scotland, with a published altitude ceiling of 4876 m and a range diameter of 3704 m. It's worth noting that this is considerably less than the minimum defined by UKRA of 7408 m for this site, indicating the uncertainty in the current guidelines, and the maximum altitude can be increased if you demonstrate that it will land within the designed range area. With detailed, validated analysis models, it would be possible to increase the current altitude limits of launch sites and to potentially propose new sites with greater capability.

Table 3.2: UKRA Minimum Launch Site Diameter

Motor Class	Total Impulse [Ns]	Diameter [m]
A	Up to 2.5	30
B	2.5 - 5	60
C	5 - 10	120
D	10 - 20	150
E, F, G	20 - 160	300
H, I, J, K, L, M	160 - 10240	450
N	10240 - 20480	600
O	20480 - 40960	900

During the launch event, the flyer and the Range Safety Officer (RSO) are responsible for ensuring the rocket is safe to fly and will not exceed the maximum altitude or launch range boundaries. This is done by ensuring the flight meets the minimum expected standard for a safe launch in the UKRA Safety Code, inspecting the rocket, and verifying the modelled flight through a deterministic simulation. This process relies on the experience/judgement of the RSO and flyer, certainty in the model inputs (wind, mass, geometry, launch angle, etc) and an accurate simulation program. The uncertainty in this process leaves room for issues to go unnoticed, which may be significant for larger, heavier rockets.

The US Code of Federal Regulations defines the minimum criteria for a commercial rocket launch based on the probability of impact with a member of the public, aircraft, water vessel or if the total risk to all members of the public does not exceed an expected value of 1×10^{-4} casualties [10]. Where a casualty (serious injury or death) is defined as an impact with 15J of kinetic energy or a blast overpressure on impact greater than 1.0 psi. Assuming a rocket has a solid motor and therefore no pressurised tanks, this allows the 15J threshold to be used with a probabilistic casualty model, alongside a stochastic trajectory and ground density model, to determine the probability of casualty and expected value. This provides a quantitative method for assessing launch risk and safety.

3.3 Weather Effects on Rocket Performance

In rocketry, weather is the one variable you can't control, yet it has a significant impact on the rocket's performance and, ultimately, its dispersion patterns. As such, it's critical to understand for accurate dispersion analysis and meeting the aims and objectives of this project. This section will focus on the different types of weather and their effects, followed by an analysis of weather forecasting and uncertainty.

Wind is the most significant weather factor affecting rocket performance. During ascent, wind causes weathercocking and affects the rocket's stability and horizontal velocity. During descent, wind affects the drift rate and ultimately the landing location. Wind speed also varies with altitude and time, resulting in distinct wind speeds and directions at different altitudes and variable-speed wind gusts. Given this significance, wind speed usually dictates whether it is safe to launch; as such, a maximum operating wind speed of 20 mph is enforced [9].

Rocket launches are usually cancelled due to **lightning** or **poor visibility**, as both pose significant safety concerns. Firstly, during lightning, it's unsafe to work with conductive launch rails (or towers) due to the risk of shock, and motors may be susceptible to accidental ignition. With a lack of visibility, it's unsafe to launch unless you can see the surrounding area and the entire area of the sky you will launch into, so that you don't injure a ground or air user. Sometimes, radar coverage is used for airspace surveillance in lieu of visual observations.

Student and amateur teams don't typically launch rockets in the **rain** due to reduced visibility, potential damage and personal preference. Whilst there is a lack of research on the impact of rain on rocket performance, Yihua Cao's work on aircraft indicates that rain leads to an increase in drag, a decrease in fin lift and can result in flight sensors failing [11].

Temperature changes affect motor thrust and air density. As the ambient temperature increases, thrust from solid motors increases because the burn rate is temperature-dependent. With hybrid and liquid motors, the relationship is more complicated and depends on the fluids, the feed system, and the engine design. Temperature increases also reduce density, which reduces thrust.

Although weather influences both rocket operation and propulsion performance, the effects of wind and temperature-derived air density will be considered within the scope of the project. The temperature-derived motor performance effect will not be considered due to the significant complexity of estimating it and the limited resources available for this project. Lightning, visibility and rain will also be considered out of scope as they don't usually affect the performance of a rocket flight in practice.

3.3.1 Weather Prediction and Uncertainty

Weather within the troposphere is primarily driven by uneven solar heating of the Earth's surface, which creates convection currents and atmospheric circulation. As air near the surface is heated, its density decreases, causing it to rise, whilst cooler and denser air sinks to replace it. These temperature differences generate pressure gradients, which in turn drive wind formation. The weather is further affected by variations in terrain and geography. Wind velocity and direction vary with altitude due to changes in atmospheric layers and air mass movement, producing wind shear. Wind shear is particularly significant in rocketry as changing wind conditions throughout ascent can induce weathercocking and increase trajectory uncertainty, whilst variable winds during descent affect drift and landing location.

Weather models produce forecasts by solving equations derived from the conservation of mass, momentum and thermal energy to predict how atmospheric variables such as temperature, pressure, humidity and wind evolve over time. To achieve this, the atmosphere is divided into a three-dimensional grid where interactions between neighbouring cells are calculated iteratively from observed initial conditions, allowing them to forecast future atmospheric states [12]. A limitation of weather models is that they depend heavily on initial conditions and the approximations used to represent the physical world. They attempt to minimise this impact by creating an ensemble of forecasts using slightly different initial conditions and models. Fundamentally weather models predict the future state of the weather model rather than the actual weather and this introduces some inaccuracies.

The Global Forecast System (GFS) is a global weather prediction model that uses a coarse grid, typically around 28 - 70 km horizontal resolution and produces forecasts for 10 – 16 days [13]. Because of its large grid spacing, the terrain is heavily simplified, with mountains and surface features averaged across each grid cell, thereby ignoring small-scale geographic effects. As a result, GFS is effective for large-scale weather patterns but less accurate at predicting localised conditions, particularly when small-scale geographic effects are significant.

Local weather models use GFS output as their initial and boundary conditions and then forecast at a much finer grid spacing of 1-3 km, which allows a more detailed representation of the surface terrain and its associated effects, improving local wind and temperature predictions. However, even at this scale, terrain is still approximated, and forecast accuracy decreases beyond short time periods due to atmospheric uncertainty. Despite finer grid spacing, local models remain constrained by parameterised terrain effects and boundary-condition dependence on GFS, limiting true local-scale accuracy [12]. For accurate local forecasts, it may be necessary to combine weather models with supplementary local data to improve forecasts.

Weather reanalysis, such as ERA5 from the Climate Data Store, uses rerun weather models after observations are made to produce a more accurate representation of the forecast state [14]. Whilst these are often more representative than weather forecasts, they are only as good as the observations used; areas with no observations may be less accurate, and they still suffer from the same issues: they are accurate over a wide area but inaccurate over small areas.

Uncertainties throughout all weather forecasts and reanalysis pose an uncertainty that ultimately leads to uncertainty in the rocket performance due to weather and the landing dispersion.

3.4 Deterministic Rocket Trajectory Models

Deterministic rocket trajectory models use physics-based equations of motion to predict rocket flight behaviour without experimental data. These models form the foundation of trajectory simulation and are essential for predicting the flight trajectory of unguided rockets. The Barrowman method provides a simplified analytical framework for estimating aerodynamic stability derivatives using small-angle approximations based on rocket geometry. 3 Degree of Freedom (DoF) models treat the rocket as a point mass and simulate translational motion under thrust, drag and gravity, whereas more advanced 6DoF models extend this by incorporating rotational dynamics and attitude changes, enabling a more complete representation of coupled aerodynamic and inertial effects during flight, such as stability margin and centre of pressure effects [15]. Simulation models are used along with the Barrowman method, motor thrust data, rocket geometry and recovery system information to simulate a flight as seen in Figure 3.3. Unmodified 3DoF are not used as they do not model stability effects.

OpenRocket is an open-source graphical application that implements a 3DoF simulation model with Barrowman-based stability analysis based on Sampo Niskanen's work [16][17]. The software is supported by technical documentation that includes limited experimental validation against real flight data, demonstrating reasonable agreement for subsonic rocket performance [18]. Its simplified aerodynamic assumptions make it particularly well suited to the design of subsonic rockets and defining geometry. In addition, its graphical user interface and intuitive design make it convenient to use.

RASAero II is a graphical rocket performance analysis application designed to provide higher-fidelity aerodynamic and trajectory prediction [19]. It incorporates experimentally derived and validated aerodynamic coefficient databases and semi-empirical drag and lift models, with particular emphasis on accurately capturing transonic and near-supersonic flight regimes where compressibility effects become significant. This makes it more suitable for rockets that exceed subsonic speeds. RASAero II has a less intuitive user interface and is usually used in conjunction with OpenRocket.

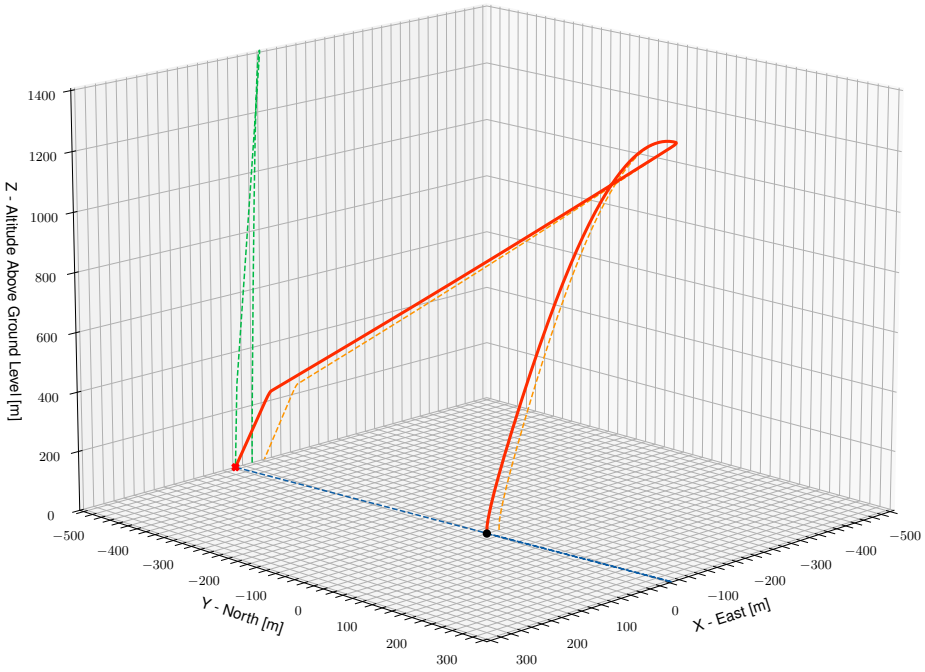


Figure 3.3: Sample trajectory of simulated flight with RocketPy.

RocketPy is an open-source Python library for rocket trajectory simulation that implements a high-fidelity 6DoF flight dynamics model [20]. It is designed to be highly extensible and scriptable, allowing all parameters to be customised, including aerodynamic drag coefficients from RASAero II (or other program or experimentally determined methods) and has been verified against 2 rocket flights [3]. Due to its extensibility, 6DoF simulation model, and extensive verification, it's the ideal choice as a basis for further research.

3.5 Stochastic Modelling Methods

Stochastic models are used to quantify uncertainty in systems where input parameters are defined by probability distributions rather than single deterministic values. In rocket trajectory analysis, this enables the uncertainty in a system to be considered, and its impact quantified in the dispersion of landing points. A stochastic Monte Carlo model addresses this limitation by repeatedly sampling input parameters from defined probability distributions and running many deterministic simulations to generate a distribution of possible trajectories. This produces a list of possible outcomes for further

analysis and confidence region determination. However, the approach is computationally intensive, and its accuracy depends strongly on the validity of the assumed input distributions, so poorly defined uncertainty can lead to misleading results [21]. Despite this, Monte Carlo methods are well-suited for rocket dispersion analysis as they can directly propagate input uncertainties into trajectory uncertainty.

Existing tools, such as the Cambridge Rocket Simulator, use stochastic Monte Carlo methods to generate dispersion patterns; however, the model does not explicitly account for non-parachute failure events [22]. Similarly, RocketPy provides a Monte Carlo simulation by integrating the deterministic flight model with a stochastic input generator, but it does not incorporate failure mode modelling [23]. A custom stochastic layer built on top of the deterministic RocketPy flight model provides the greatest flexibility for defining input uncertainty and extending the model to include failure conditions. This provides the most suitable approach to meet this project's aims.

3.6 Visualisation of Dispersion and Risk

Visualising rocket dispersion and risk is critical for communicating the output of dispersion analysis in a clear, understandable way. This section works towards the aim whilst assessing visualisation approaches. A dispersion analysis plot shows the distribution of possible rocket landing positions resulting from input uncertainty. Instead of plotting a single ideal landing point, a cluster of landing points is plotted to represent the probable landing location in the uncertain system. This is important because a single landing point can be misleading and doesn't reflect the system's uncertainty.

Circular Error Probable (CEP) is a simplified statistical measure that defines the radius within which 50% of landing outcomes are expected to occur, assuming a circular distribution of error centred on a mean impact point [24]. It is widely used due to simplicity; however, it assumes isotropic uncertainty, which does not reflect realistic rocket dispersion patterns. Covariance ellipsoids represent the uncertainty in multiple dimensions, producing an elongated shape that better reflects anisotropic error distributions [25]. This makes them more suitable for rocket trajectory analysis, but at the cost of greater computational complexity. These methods provide an effective way to display dispersion-trajectory confidence regions.

Visualising risk is critical for communicating the significance of each probable landing point and its impact. Impact energy can be plotted directly to compare the impact of each area, or an energy threshold can be used to determine which areas would result in casualties [10]. This can be extended to a casualty or impact heatmap by integrating population density data to directly compute and visualise the risk to people, and similarly can be applied to vehicles and property. Together, these visualisations enable a more complete assessment of rocket risk by combining probability of impact

with potential severity.

3.7 Validation and Flight Computer

Model validation is a critical component of this project as it directly supports the aim of developing and evaluating a reliable trajectory and dispersion prediction model for suborbital unguided rockets. A flight computer with data logging capability would be a useful addition to an experimental model validation approach, providing more information to aid decision-making.

Accurate trajectory reconstruction relies on onboard sensors, primarily GPS, IMU, and barometric pressure sensors. GPS provides global position estimates but is typically limited to metre-level accuracy and can suffer from signal dropout during high-dynamic flight [26]. The IMU measures acceleration and angular rates but is subject to drift when integrated to estimate position over time. The barometer provides altitude estimates based on atmospheric pressure; however, it relies on consistent ambient air pressure.

To address these limitations, sensor fusion and data processing techniques are required. Time synchronisation between sensors is essential to ensure consistent state estimation across all data streams. GPS data must be parsed and converted into position and velocity estimates within a consistent coordinate frame. Barometric altitude is derived using the International Standard Atmosphere (ISA). A Kalman filter may be used for sensor fusion and state estimation; however, this level of complexity may be unnecessary if GPS and barometric data already provide sufficient accuracy for dispersion-level validation rather than full high-fidelity trajectory reconstruction [27].

Validation is performed by comparing real flight trajectory with simulation outputs, focusing on key metrics such as landing position, apogee and dispersion behaviour. Results should be benchmarked against commercially available certified devices, as these provide an accurate reference for assessing whether the model produces realistic flight behaviour.

4 Methodology

To determine the probable landing location, the model aims to incorporate as many sources of uncertainty as possible within the project's resource constraints. This starts by defining sources of uncertainty into 5 broad categories:

- **Environmental Uncertainty** is defined as wind and weather uncertainty.
- **Flight Event Uncertainty** arises from flight events such as parachutes exhibiting unexpected performance, such as a failure, and from the other failure modes identified in the NAR safety report [8], and with associated probabilities in Table 3.1.
- **Design and Manufacturing Uncertainty** relates to design decisions and the accuracy of manufacturing the system according to the design.
- **Launch Uncertainty** relates to launching the rocket, including the launching hardware, inclination and heading.
- **Performance Uncertainty** is the variation in performance of commercial/certified products used in rockets, such as motors and parachutes. Critically, this includes the system's actual performance rather than its advertised design point.

Section 4.1, shows the input parameters, their associated uncertainty category, the deterministic design point for each parameter and the uncertainty distribution to stochastically represent each input. Within this table, the rocket's nose tip is taken as the reference point.

This model assumes that all input parameters' performance follows a normal distribution, with the mean at the design point and the provided variance. Where the variable is provided as a percentage, it is calculated by multiplying the percentage by the mean. This enables the distribution to reflect the object's scale, as manufacturers' workshops define tolerances according to the ISO 2768 standard [28]. The distribution of most parameters is defined by a fixed variable, such as "Performance Uncertainty," which allows the uncertainties of all performance parameters to be changed simultaneously to represent different capabilities and use cases, similar to ISO 2768's tolerance classes.

For simplicity and due to the difficulty in sourcing performance data, the distribution of uncertainty in this model has been estimated by the author, based on experience with high-powered rocketry in the UK. Where other sources are readily known and have been used, this has been stated. Nonetheless, it is likely that this model provides a suitably accurate representation of actual rocket flights, as will be assessed in the results, and can be used to assess performance, perform dispersion analysis, and derive relationships, significance, and trends. This represents an area for future work to extend this method further.

Table 4.1: Summary of case study rockets and uncertainty.

Parameter	Units	Validation Rocket	UK High Altitude	Student Outreach
Summary				
Length	m	0.76	2.4	0.67
Mass	g	762	13435	562
Diameter	mm	72.4	102	56.3
Motor		F31-12A	M2020-P	E30-7
Total Impulse	Ns	55.5	9429	33.6
Apogee	m	169	5286	172
Input Variance				
Launch Rail Length	m	0.1	0.1	0.1
Launch Inclination	deg	4	2	4
Launch Heading	deg	10	10	10
Wind Speed	m/s	1.5	1	1
Wind Heading	deg	10	10	10
Performance Uncertainty	%	1	1	1
Design and Manufacturing Uncertainty	%	2.5	2.5	5

4.1 Case Studies

This project uses three case studies for analysis throughout and to represent the different types of rockets that could benefit from improvements in dispersion analysis. Section 4 provides an overview of these rockets and their flights, and Section 4.1 defines the design points for each case study, along with their associated distributions.

The validation rocket, depicted in Figure 4.1 and Figure 4.5a, is the modification of a flight-proven mid-power rocket to carry a data-logging flight computer to record trajectory data throughout the validation flight. Note the apparent damage on the fin in Figure 4.5a is purely cosmetic and has no impact on the outcome of the flight.

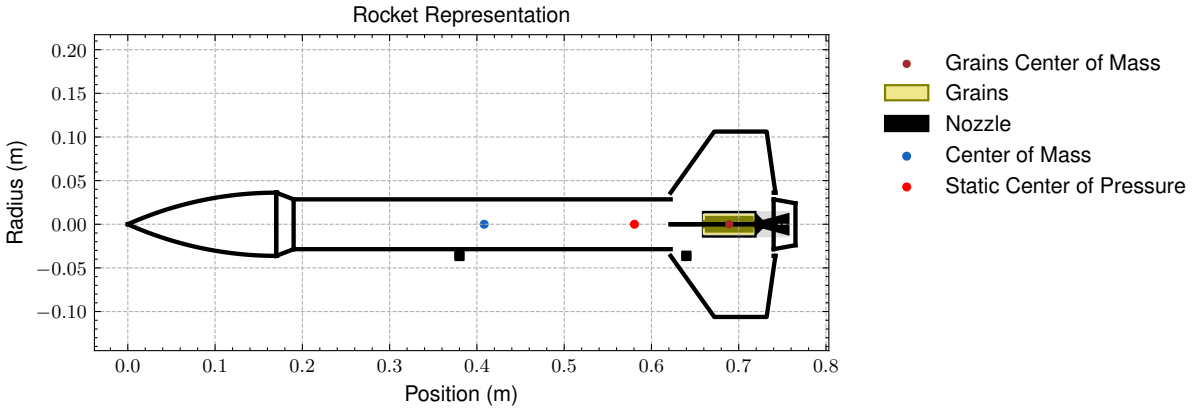


Figure 4.1: Diagram of the Validation Rocket.

The UK High Altitude rocket, depicted in Figure 4.2, represents one of the largest single-stage rockets that can be launched from a UKRA-accredited launch site within the UK and is typical of rockets that many university rocketry teams are building to launch within the UK.

Table 4.2: Case study data source and corresponding uncertainty category defintions applied in this study.

Parameter	Units	Validation Rocket	UK High Altitude	Student Outreach	Uncertainty Distribution
Launch Uncertainty					
Launch Rail Length	m	1.8	4.6	1.5	0.1 m
Launch Rail Inclination	deg	85	90	90	4 deg
Launch Rail Heading	deg	60	176	231	10 deg
Environmental Uncertainty					
Wind Speed	m/s	[a]	4.5	5	1 m/s
Wind Heading	deg	[a]	176	51	10 deg
Performance Uncertainty					
Main Parachute Diameter	mm	457	1524	500	[b]
Main Parachute Drag Coefficient		0.8	2.2	0.8	[b]
Main Parachute Trigger Altitude	m	N/A	Apogee	N/A	[b]
Drogue Parachute Diameter	mm	N/A	609.6	N/A	[b]
Drogue Parachute Drag Coefficient		N/A	1.6	N/A	[b]
Drogue Parachute Trigger Altitude	m	N/A	300	N/A	[b]
Motor Ejection Delay	s	7	N/A	7	[c]
Motor Diameter	mm	29	75	24	[b]
Motor Length	mm	98	893	70	[b]
Motor Mass	g	102	7032	47	[b]
Motor Propellant Mass	g	70	4349	18	[b]
Motor Burn Time	s	1.8	4.2	1	[b]
Design and Manufacturing Uncertainty					
Motor Position	mm	659	1398	600	[d]
Motor Thrust Misalignment	mm	0	0	0	[d]
Rocket Diameter	mm	72.4	102	56.3	[d]
Rocket Mass (excl motor)	g	659.5	6403	515	[d]
Rocket Inertia (x)	kgm^2	0.000469	0.01408	0.00147	[d]
Rocket Inertia (y)	kgm^2	0.000469	0.01408	0.00147	[d]
Rocket Inertia (z)	kgm^2	0.029286	2.326591	0.014072	[d]
Rocket Centre of Mass	mm	365	1124	320	[d]
Rail Button Lower Position	mm	380	1550	305	[d]
Rail Button Upper Position	mm	640	2050	611	[d]
Nose Length	mm	170	500	120	[d]
Fins Root Chord	mm	120	171	75	[d]
Fins Tip Chord	mm	60	51	40	[d]
Fins Span	mm	70	120	43	[d]
Fins Sweep Length	mm	50	120	28	[d]
Fins Position	mm	622	2079	585	[d]
Tail Aft Diameter	mm	480	83	N/A	[d]
Tail Length	mm	25	56.7	N/A	[d]

[a] Wind speed and heading are defined by either fixed values or related to altitude.
[b] Performance Uncertainty defined as a percentage variance of the design value.
[c] Maximum Motor Certification Uncertainty defined by NAR motor certification standards [29].
[d] Design and Manufacturing Uncertainty defined as a percentage variance of the design value.

The Student Outreach rocket is a rocket designed and built by students (ages 11-18) participating in the national UKROC competition. This rocket, as depicted in Figure 4.3, may be used to assess whether it would be safe for a student rocketry team to launch from their own launch site, such as a large playground or open field. Including this model also enables a sensitivity analysis comparing large and small rockets.

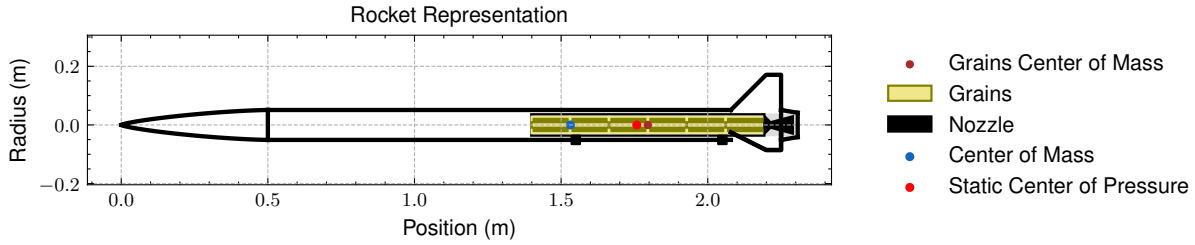


Figure 4.2: Diagram of the UK High Altitude case study rocket.

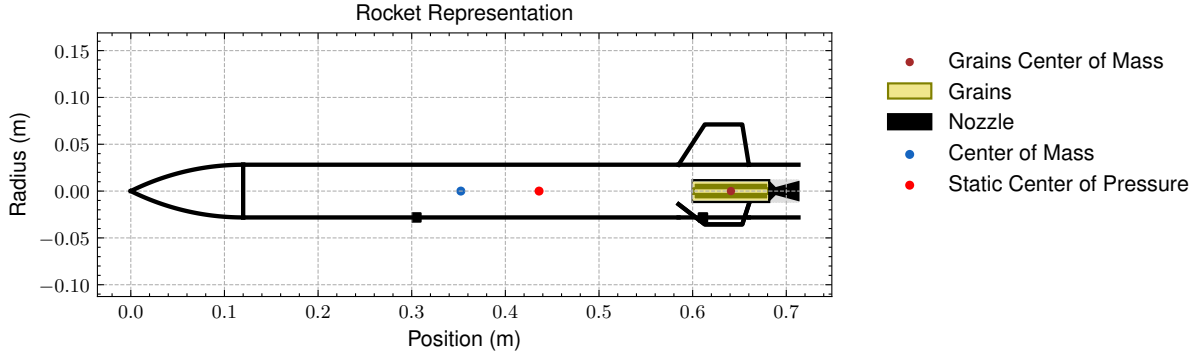


Figure 4.3: Diagram of the Student Outreach case study rocket.

4.2 Dispersion Model

A stochastic Monte Carlo model implemented based on the RocketPy 6DoF trajectory model has been developed, supporting stochastic inputs and the simulation of failure events. Figure 4.4 illustrates the data flow through this model and can be broken down into three distinct sections: data ingestion and preprocessing, Monte Carlo model with failure event logic, and post-processing and visualisation.

The data ingestion and preprocessing stage involves calculating or defining all deterministic parameters for the Monte Carlo model and then assigning stochastic distributions to those parameters if they are sampled within the core model. This starts by defining the rocket's design below, using it to calculate the key parameters needed for the deterministic simulation model to work, collecting the datasets needed by the model, and then assigning stochastic distributions to relevant parameters.

- The rocket geometry and mass distribution were defined within OpenRocket to establish the baseline physical configuration of the vehicle.
- The inertia properties of the rocket were estimated with OpenRocket using the methodology described in Appendix A.2.
- Aerodynamic performance characteristics were computed using RASAero II, with results processed via a data extraction script as described in Appendix A.1.
- Lift curve slope data for the fin airfoil was obtained from the UIUC Airfoil Database, or alternatively, approximated using flat-plate assumptions [30].
- Failure probability data was incorporated from the dataset presented in Table 3.1 to

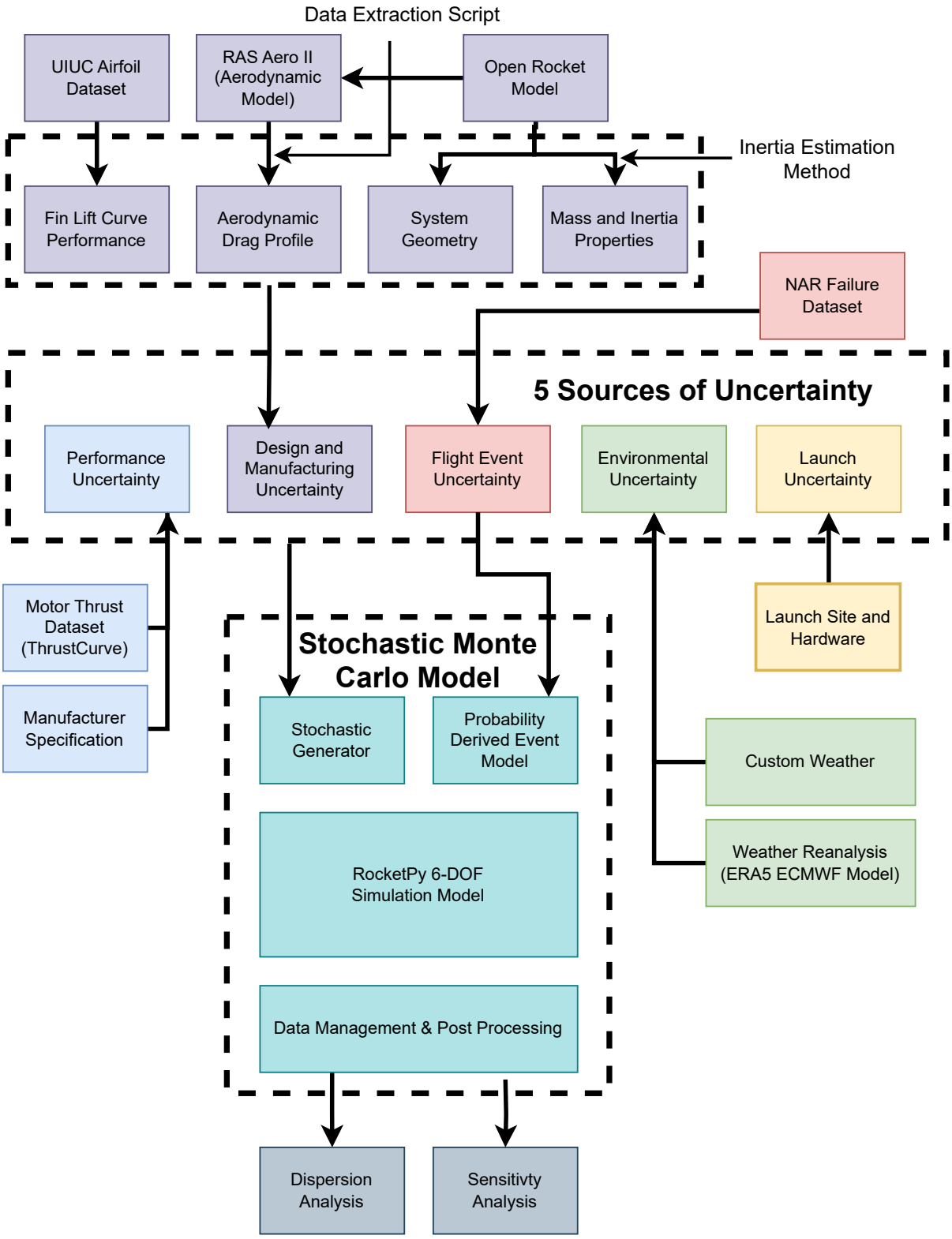


Figure 4.4: Schematic of data flow from input preparation to outputs.

estimate stochastic failure events within the simulation.

- Motor thrust curves were obtained from ThrustCurve.org for the selected motor, and corresponding motor parameters (dimensions, burn time and total impulse) were defined accordingly [31].

- Atmospheric conditions were derived from ERA5 reanalysis pressure-level datasets (netCDF format) or alternatively specified using defined wind speed and direction profiles at the launch site [14].
- Launch rail configuration parameters were defined by the physical launch site setup and incorporated into the simulation environment.
- Stochastic input distributions for all uncertain parameters were defined in accordance with Section 4, with variance and distribution derived from baseline parameter values listed in Table Section 4.1.

The second stage includes the Monte Carlo model and failure simulation. The computation in this stage is repeated 5000 times, resampling each time to produce a stochastic output that reflects the model's input parameters. This process starts by sampling all the parameters from the previous stage to create a simulation starting state. Based on the probabilities defined in Table 3.1, the failure mode for each simulation is determined. A minimum event probability of 0.5% is enforced to ensure that low-likelihood events are still represented in the simulation, even where historical trends suggest they are statistically unlikely to occur. This assumption is considered appropriate because the potential consequences to people or property outweigh the conservatism it introduces. The failure modes are then simulated, making use of the existing RocketPy tools and equivalent physical systems to approximately simulate the failures as below:

- **Nominal** failure mode simulates the flight without any failures.
- **Unstable** failure is not considered, as this is already simulated by RocketPy through the 6DoF model.
- **Lawn Dart** failures where the rocket doesn't deploy its parachute and assumes a ballistic flight are modelled by not deploying the parachute.
- **Core Sample** failures, which represent a ballistic flight without a nose cone, are simulated by deploying a parachute with drag equivalent to the rocket's body cross-section.
- **No Chute**, where no parachute is deployed, but the rocket is not ballistic, is simulated with a parachute with the equivalent drag of the rocket body modelled as a cylinder tumbling.
- **Separation** failure modes, where the rocket free-falls from the parachute deployment point, are modelled as a parachute with a very small amount of drag.
- **Shred**, where the rocket disintegrates, is modelled by placing a sudden, very high drag air brake device towards the end of the motor burn, as this is usually the point of maximum dynamic pressure for a rocket flight.
- **Motor CATO**, is modelled by suddenly halting rocket thrust whilst the rocket is burning. It's worth noting that this doesn't consider the blast hazard to people and property, although, on balance, if following the UKRA Safety Code, people should be

outside the blast radius [32].

After the simulation's starting state is modified to model potential failure events, the RocketPy 6DoF trajectory model is run to simulate the flight. Useful outputs, such as the apogee, landing position, and impact energy, are saved to support future analysis.

Finally, the data is processed to calculate variance ellipsoids that represent the probable landing location of the modelled rocket. The model's output is then plotted as ellipsoids, raw data, and satellite maps to provide a contextual view of the rocket's dispersion pattern.

The implementation source code for this model is available at, `https://github.com/rufusclark/Aerospace-Toolbox/`

4.3 Experimental Validation

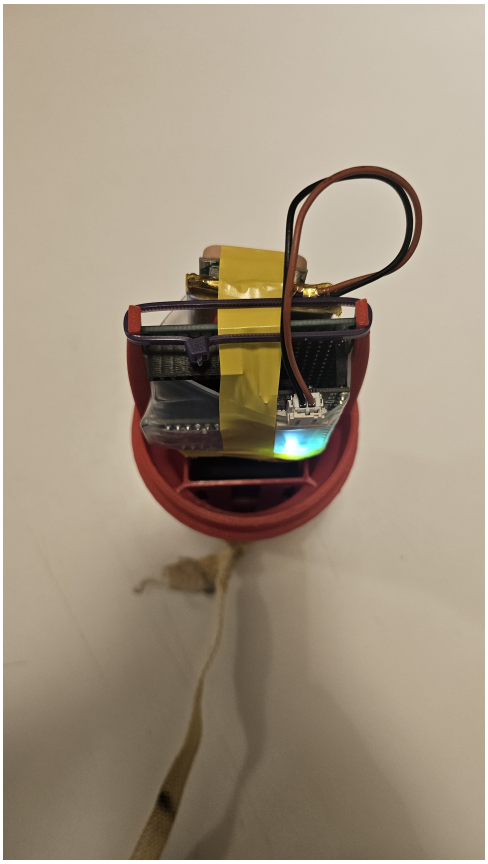
For this method to be widely applied and for its outputs to be trusted, it should be experimentally validated through actual rocket flights. This comprises 2 parts: validation of the flight computer's accuracy and validation of the dispersion analysis' accuracy, both assessed against known good values. Ideally, at least 2 flights would be conducted under different conditions to reduce random error and the impact of anomalies, but in this case, a single flight with the validation rocket case study will be conducted due to this project's resource constraints.

During model validation, the launch point, launch heading, launch angle, landing point, and flight video will be recorded, and a dispersion analysis will be conducted using weather reanalysis to retrospectively assess the simulation's accuracy. The launch and landing points will be measured with a smartphone, ensuring that GPS horizontal accuracy is maximised before each measurement. The launch heading will be estimated from the headings of known fixed objects in Satellite Maps, and the launch inclination will be measured with a digital angle measure. The video will be used if needed to investigate any anomalies that arise. This experiment will be considered a success if the rocket lands within the 2σ horizontal covariance ellipsoid that represents 86.5% of all flights.

Secondly, the flight computer itself, as shown in Figure 4.5b, will be validated to ensure it records data that is representative of the actual flight trajectory. The GPS sensor will be validated against the launch and landing points recorded for the previous section. The barometer and GPS altitude will be compared against the apogee measured by the commercial altimeter, Jolly Logics Altimeter Two, which is certified for use in rocketry competitions and claims to be precise to the nearest meter [33]. If all data sources are consistent, the flight computer will be considered sufficiently accurate for further



(a) Rocket.



(b) Payload

Figure 4.5: Flight hardware for the Validation Rocket.

data analysis.

4.4 Sensitivity Analysis

Using the Monte Carlo method, a sensitivity analysis will be conducted on both the UK high-altitude and student outreach case studies by independently varying uncertainty input categories using the parameters defined in Section 4 and Section 4.1. The resulting effects on trajectory behaviour and landing dispersion will then be analysed to identify the dominant contributors to prediction uncertainty and assess the robustness of the simulation model.

5 Results and Discussion

The section contains the results and critical discussion of the model validation and sensitivity analyses, followed by additional analyses, all intended to meet this project's aims and objectives and provide a conclusion.

5.1 Experimental Validation

Figure 5.1 compares the altitude measurements obtained from the GPS and pressure sensor throughout the validation flight, along with the measured apogee of the commercial altimeters. All sensors indicated a very similar apogee. The GPS data lagged behind the pressure data during the ascent phase but otherwise showed strong agreement after apogee. This indicates that this GPS performs poorly in high-acceleration environments but shows no similar issues on the gentler descent, consistent with the literature reviewed. These results indicate that GPS and pressure altitude are likely representative of actual flight and are suitable for further analysis. However, GPS altitude should not be used under high acceleration.

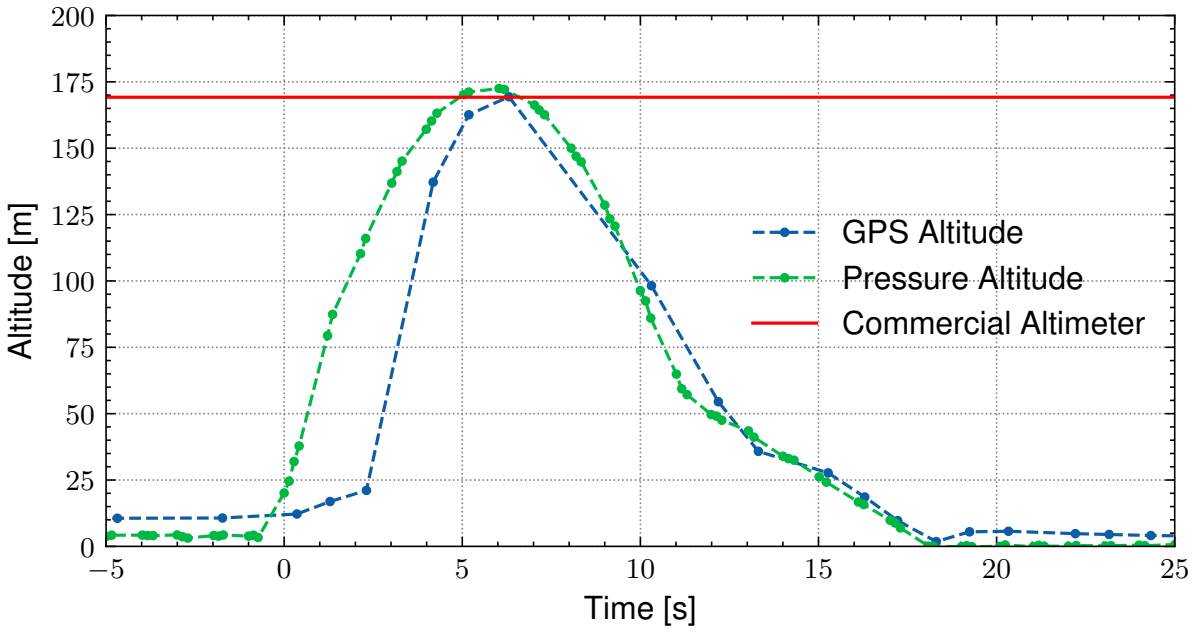


Figure 5.1: Comparison of the Validation flight onboard GPS and barometer (pressure) altitude with the commercial altimeter.

Figure 5.2 shows wind speed and wind heading versus elevation for the validation flight, based on reanalysis of the weather using the ERA5 pressure-level dataset. This shows that both the wind speed and wind heading vary significantly at the elevations through which the rocket flew. Between the rocket's launch and apogee at 169.2 m, the modelled wind speed increases from 4 m/s to 11 m/s (1 m/s increase every 24 m), indicating a strong pressure of wind shear. Likewise, the wind heading also changed

by 10 degrees in this time. From this, you can presume that assuming a constant wind speed and heading is inaccurate and unrepresentative of actual wind behaviour, even at relatively low altitudes. This conflicts with the tools available for many simulators, such as OpenRocket, which currently lack this feature.

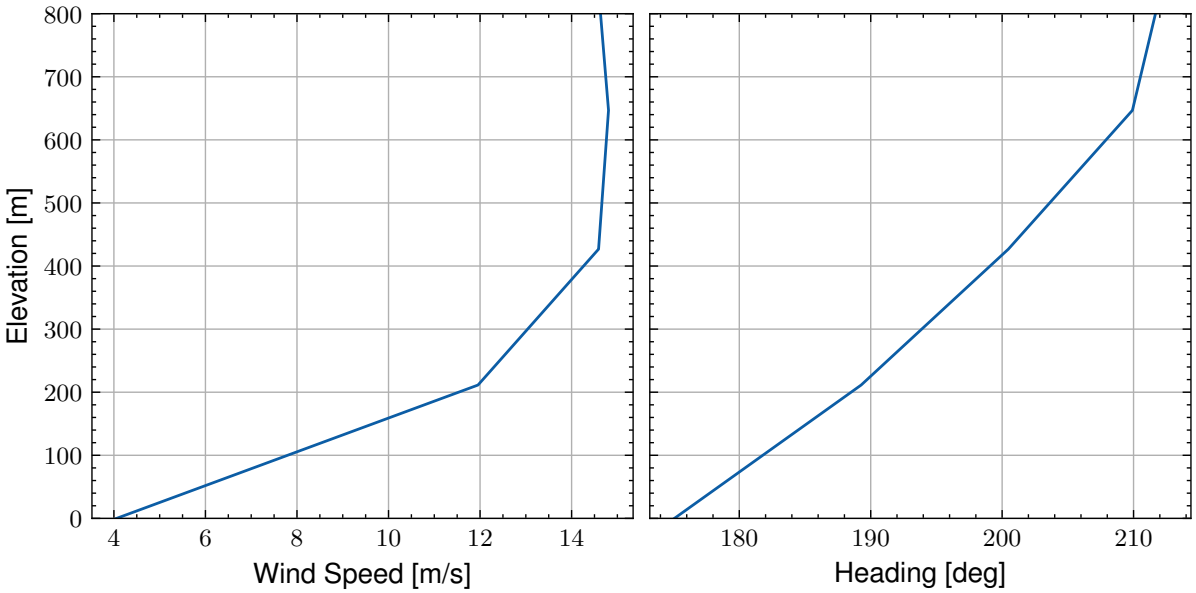


Figure 5.2: Wind speed and heading for the Validation flight derived from ERA5 weather model reanalysis.

Figure 5.3 is the dispersion analysis of the validation flight using the Figure 5.2 wind speed and heading data from the ERA5 weather model reanalysis. This shows the calculated positions of 5000 sampled outputs, labelled by their failure mode; the calculated covariance ellipsoids representing the distribution of landing points; the wind and launch heading, as annotated; and the launch and landing points. The experimental landing point is observed to lie within the 1σ covariance ellipsoid, which contains approximately 39.3% of all simulated landing points. This suggests that, for this case, the Monte Carlo model produces a statistically significant representation of the system. However, this validation is limited to a single flight case. While the results indicate agreement between the model and experimental observations, it does not eliminate the possibility that this case is statistically favourable or an outlier. Therefore, although the model shows promising agreement in this instance, further validation across multiple flights under different conditions would be required to establish broader model reliability. Within the limited scope of this project, this will be considered validation, alongside these limitations, as it meets the project objectives of developing a simulation model to predict flight trajectories and validating the model using experimental data.

Additionally, the validation of the model during nominal flights does not validate the model against non-nominal failure events. Due to the lack of available data for these

failure events and all simulated failures flights exhibiting reasonable landing locations in Figure 5.3 with theory consistent with the literature in Section 3.1.2, the flight failure event probability model will be assumed to be representative.

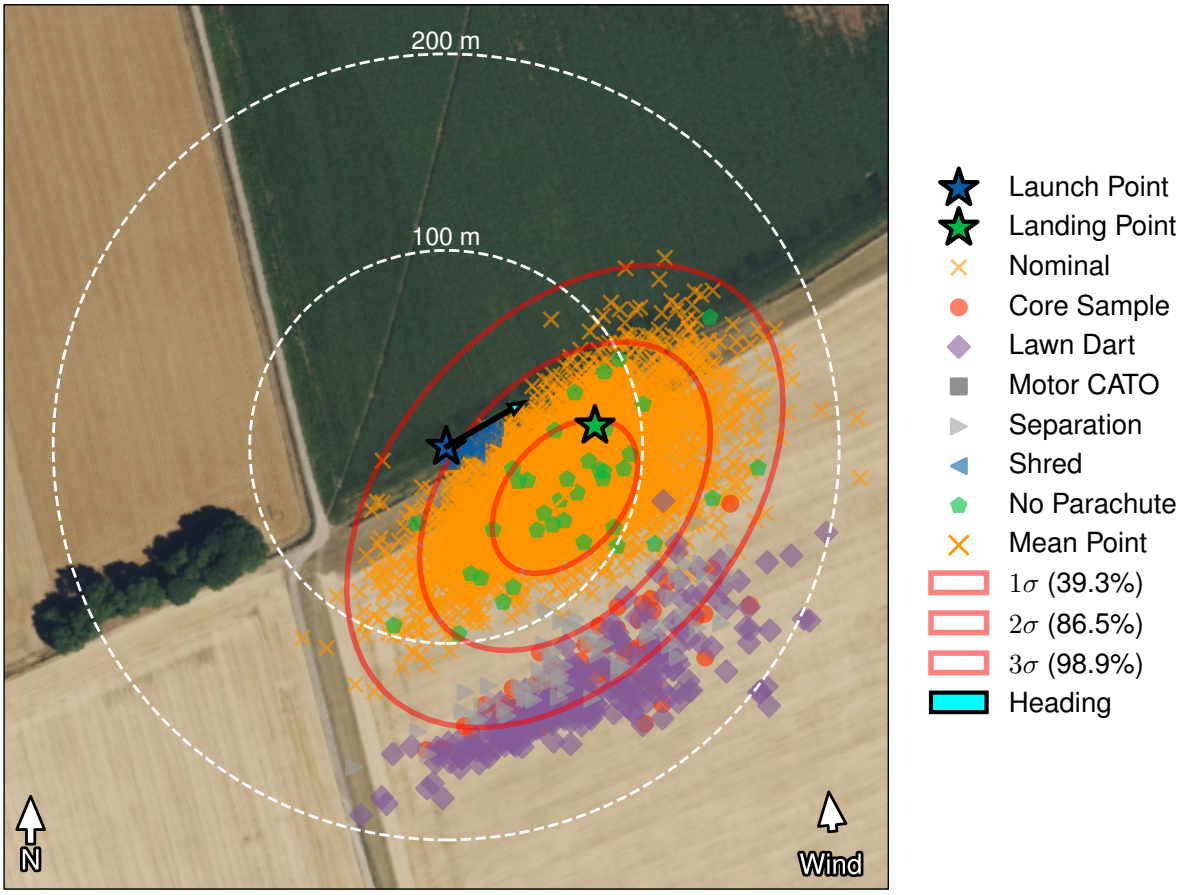


Figure 5.3: Dispersion analysis of the Validation flight using ERA5 reanalysis wind sounding.

5.2 Sensitivity Analysis

This section presents and discusses the results from 2 separate sensitivity analyses conducted for the UK high-altitude and student outreach case studies, followed by a brief overview of the analyses.

Figure 5.4, shows a set of 6 dispersion analyses based on the UK high-altitude case studies defined in Section 4.1. Each subfigure represents the simulated landing location of 5000 sampled flights, where only one of the five uncertainty categories has been varied, and all categories have been varied in the combined case. Each subplot contains its own covariance ellipsoids to represent the landing dispersion, the launch point, and the distribution's centre. Each failure mode is identified by the shape in the key. Additionally, the landing points are colour-coded on a logarithmic scale to quantify each point's impact, with the key colour representing the maximum impact energy throughout the figure. Additionally, Table 5.1 provides a summary of this sens-

itivity analysis by showing, for each case, the distance between the ideal landing point (no variations) and the mean point, the size of the 3σ covariance ellipsoids in m^2, and the equivalent contribution to this area from each category. Within this table, the contribution percentage represents an approximate measure of sensitivity and can identify which factors should be investigated to reduce system-level uncertainty.

Within Figure 5.4, environmental variation is by far the largest contributor to the size of the dispersion area and has the widest (tangent to wind direction) distribution of landing points. Table 5.1 quantifies the environmental variation contributing to dispersion area as approximately 94.9%. This can be attributed to the significant impact wind has on high-altitude flights with very long descents, as well as the relatively large uncertainty in both wind heading and speed. This indicates that additional work is necessary to develop methods to reduce environmental uncertainty and, thereby, the overall landing dispersion uncertainty. Separately, launch uncertainty is the second-largest contributor to dispersion at 4.9% and is the most sensitive factor the launch team can control. As such, minimising launch uncertainty is advisable to reduce controllable dispersion.

Figure 5.4 also shows that flight event variation is by far the largest contributor to high-energy impacts, with lawn darts, core samples, and separation events being the highest-energy events, with impact energies above 100 kJ, far exceeding the casualty threshold of 15J. In the combined simulation, these events are clustered around an ellipse of approximately 3 x 1km that falls mostly outside of the 3σ ellipsoid. This represents the area that very high-energy impacts falling outside the region which contains 98.9% of all landings. All efforts should be made to minimise the probability of these failures occurring and minimise the risk they pose by removing people and property from this area. This represents a dangerous situation in which conventional rocket dispersion methods without failure-event modelling may miss these events entirely. Whilst the over 100 kJ impacts would pose a hazard, further analysis is needed to determine whether the 15J threshold is suitable for amateur rocketry. In this simulation, all landings exceeded this threshold, even those descending within the recommended speeds of 5–8 m/s, considered safe by UKRA [34]. Overall, the results suggest that failure-event modelling provides additional insights into extreme-impact scenarios that are not adequately represented by nominal trajectory dispersion analysis alone.

Similarly to Figure 5.4 and Table 5.1, which represent the sensitivity analysis of the UK high-altitude case study, Figure 5.5 and Table 5.2 represent the sensitivity analysis of the student outreach case study using the same definitions and descriptions.

Table 5.2 shows that launch variation (96%) and environmental variation (3.5%) are the two most significant categories that influence dispersion area. This is attributed to the flight having a much shorter flight time; as a result, the landing area is more influenced by the starting inclination and heading than by wind heading and speed,

Table 5.1: Sensitivity analysis summary of independent impact of uncertainty groups on dispersion using the UK High Altitude case study.

Uncertainty	Distance [m]	3σ area [m^2]	Contribution [%]
Environment	52.8	8101100.4	94.887
Launch	8.6	419444.4	4.913
Performance	0.2	41.0	0.000
Flight Event	233.3	16603.6	0.194
Design and Manufacturing	0.6	464.1	0.005
Combined	268.1	13415084.5	

because the wind has less time to influence the flight. This indicates that efforts to reduce launch uncertainty, such as verifying the launch rail inclination, are likely to reduce the dispersion area.

Figure 5.5, shows high-energy events above 100J scattered throughout the launching area but are largely upwind of the launch point. This dispersion pattern is attributed to the highest-energy failure events, which result in ballistic flights that weathercock into the wind after apogee. The hazard associated with this can be minimised by ensuring people and property are not located upwind of the launch pad within the launch range.

Table 5.2: Sensitivity analysis summary of independent impact of uncertainty groups on dispersion using the Student Outreach case study.

Uncertainty	Distance [m]	3σ area [m^2]	Contribution [%]
Environment	0.1	161.1	3.543
Launch	0.4	4374.3	96.227
Performance	0.4	0.5	0.010
Flight Event	4.5	3.0	0.065
Design and Manufacturing	2.0	7.0	0.154
Combined	1.3	9630.7	

By comparing the sensitivity analyses of the two case studies, it can be inferred that launch uncertainty is the dominant contributor to dispersion uncertainty at lower apogees, whereas environmental uncertainty becomes dominant at higher apogees. Across both analyses, dispersion uncertainty may be reduced by minimising input-parameter uncertainty, with the most sensitive parameters yielding the greatest reduction in output variance and therefore representing the most effective targets for mitigation. However, such reductions must always be considered alongside impact energy and failure probability, with efforts focused on minimising both the likelihood and consequence of hazardous events. It should also be noted that the model is only representative when it is accurately defined; otherwise, its outputs may become inaccurate or misleading. This is particularly important for wind speed and heading inputs, given their inherent uncertainty and strong influence on trajectory dispersion.

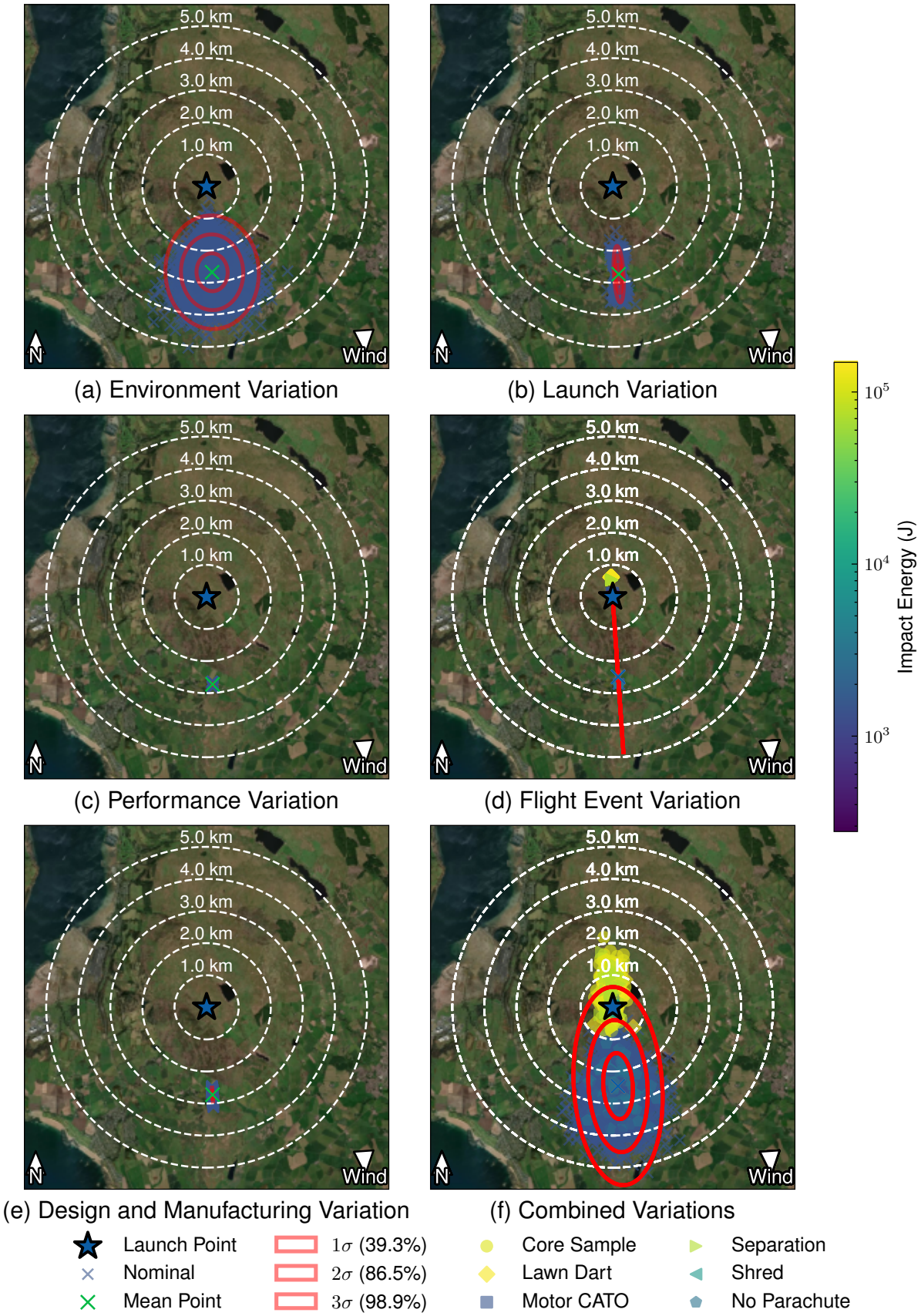


Figure 5.4: Sensitivity analysis of independent impact of uncertainty groups on dispersion using the UK High Altitude case study.

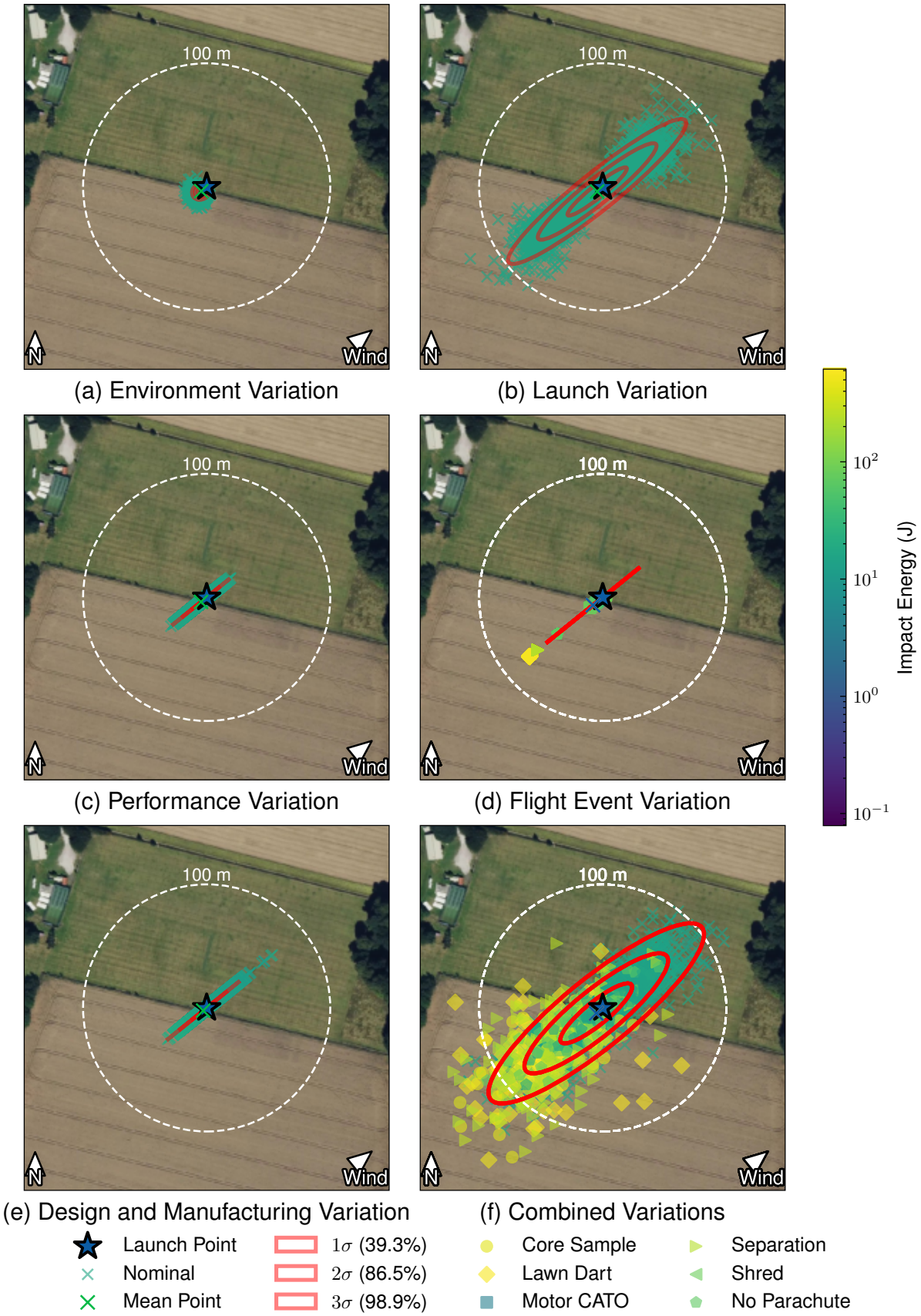


Figure 5.5: Sensitivity analysis of independent impact of uncertainty groups on dispersion using the Student Outreach case study.

5.3 Weather Prediction

This section contains additional analysis that builds on the experimental validation of the model and the conclusions from the sensitivity model. This provides a proof of concept that aims to reduce weather uncertainty by estimating wind speed and heading from flight data and then using this data for a dispersion analysis.

Figure 5.6 shows the horizontal speed and heading of the validation flight derived from GPS and barometer data. This data shows that after the parachute is deployed at approximately 50 m on descent, the wind speed and heading stabilise, indicating the rocket is now drifting with the wind. Assuming that all of the horizontal speed is attributed to the wind, the horizontal speed and heading can be determined between 10 and 30 m as 5.2 m/s and 148 degrees, respectively. Then, assuming the wind speed and header are not dependent on elevation, they are treated as fixed values for the entire flight.

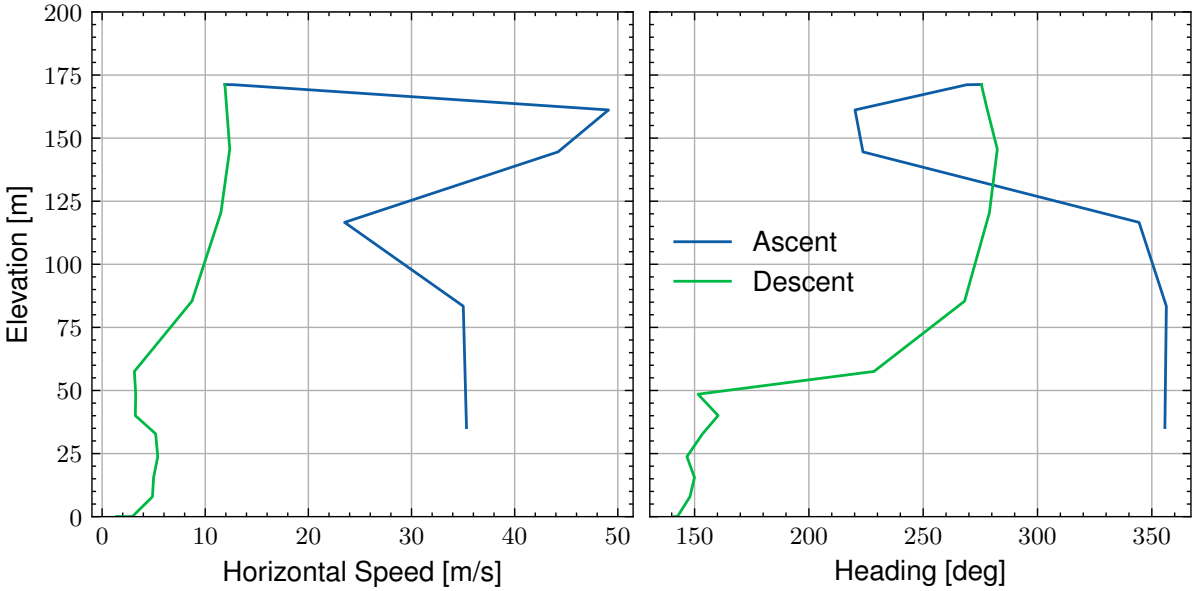


Figure 5.6: Horizontal velocity and heading for the Validation flight derived from GPS and barometer data.

Figure 5.7 is the dispersion analysis of the validation flight using the estimated wind speed and header. This shows the calculated positions of 5000 sampled outputs, labelled by their failure mode; the calculated covariance ellipsoids representing the distribution of landing points; the wind and launch heading, as annotated; and the launch and landing points. The experimental landing point is observed to lie within the 2σ covariance ellipsoid, which contains approximately 86.5% of all simulated landing points. This suggests that, in this case, the Monte Carlo model produces a statistically consistent representation of the system, and that this approach may be valid, although more flights would be needed to prove it for certain.

This proof-of-concept demonstrates that this technique works to feed back the estimated wind speed and heading into the same rocket flight. Further work would be needed to improve this technique by refining the underlying data using a sensor-fusion algorithm that combines IMU, barometer, and GPS data, and by accounting for horizontal speed not due to wind. These results indicate that it may be possible to use the data from one flight to reduce the environmental uncertainty in a subsequent flight, ultimately reducing the dispersion uncertainty. This assumes the wind speed and heading will not change significantly between the two flights.

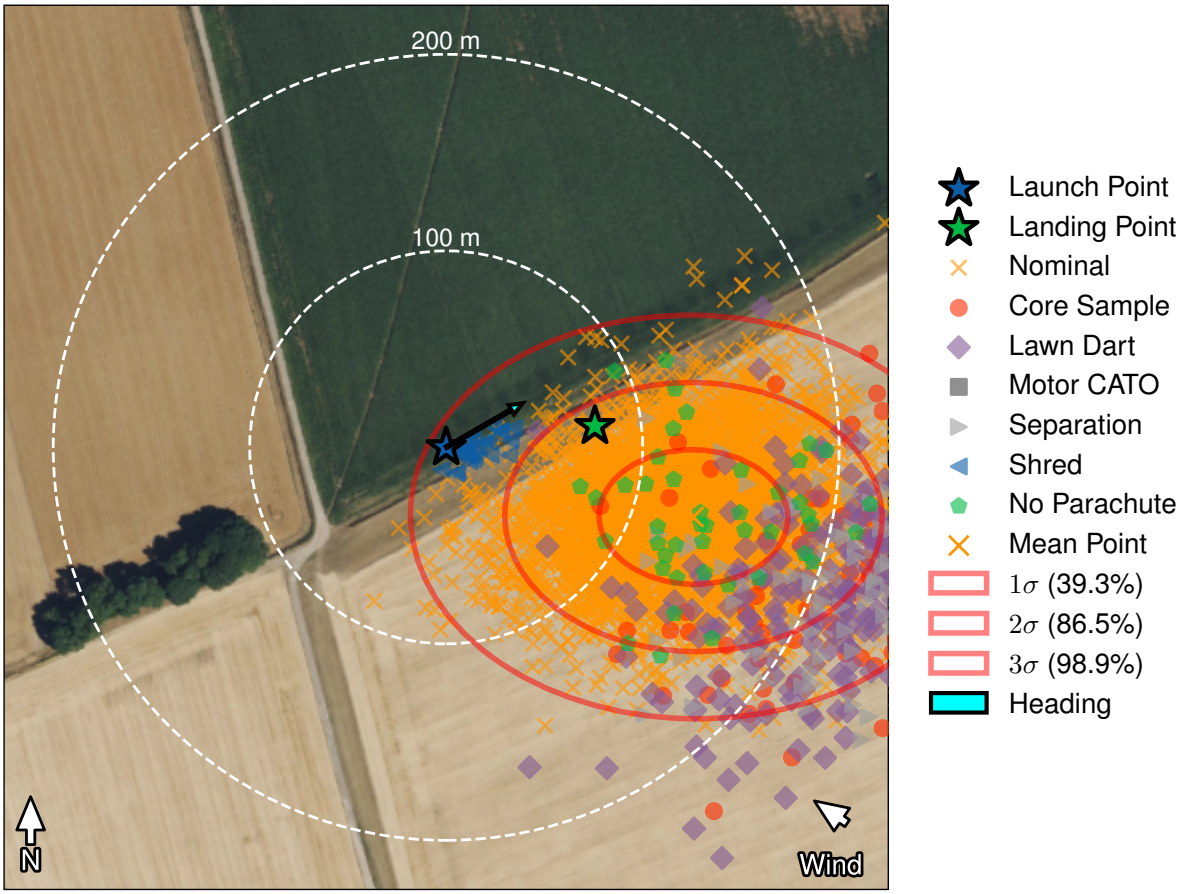


Figure 5.7: Dispersion analysis of the Validation flight using ERA5 reanalysis wind sounding.

6 Impact Assessment

This project focuses on predicting the landing location of rockets used by student rocketry teams, but is applicable to any suborbital unguided rocket, including hobby rockets and those used in STEM competitions for students. Throughout this project, consideration has been given to environmental, societal and ethical impact, as part of the project planning and decision-making process.

Environmental Impact

The environmental impact was considered throughout design and testing. Estimated emissions include:

- Transport: 44160 gCO2e (320 km return to closest launch site)
- Laptop and/Monitor: 2,745 gCO2e (300 h at 150W)
- Soldering: 30.5 gCO2e (5 h at 100W)
- Simulation: 164.64 gCO2e

Material impact was minimised by reusing an existing rocket airframe and borrowing payload components. Only rocket motors were purchased specifically for the project. Overall, the largest contributors to emissions were transport and prolonged computation rather than manufacturing, highlighting the importance of efficient testing strategies and remote simulation where possible.

Societal Impact

This project has already contributed to the safety case for a school rocket launch event in the wider Manchester area, enabling a launch that would otherwise be difficult to risk assess. There is also emerging interest in the approach from UKRA, including the potential to use this work to assess both rockets and launch sites.

Ethical Considerations

Ethical considerations focus primarily on improving safety in model rocketry and reducing the risk of harm to participants and spectators. The system supports more informed launch decisions, ensuring rockets land within safe, predicted areas.

Conclusion

Overall, the project demonstrates a net positive impact across environmental, societal, and ethical dimensions, with opportunities for further emission reduction and wider adoption in safe student rocketry practices.

7 Conclusion

This project investigated the dispersion of unguided sub-orbital rockets by developing a stochastic Monte Carlo model that predicts the landing distribution of rockets while simulating flight failure events based on historical failure data and defined input parameters and uncertainty.

The developed model was validated using experimental rocket flight data. The results showed that the actual landing was within the predicted 39.3% confidence ellipses, indicating that the model offers a realistic representation of real-world flight. This demonstrates the model's reliability and suitability for use in landing dispersion and further analysis.

The sensitivity analysis showed that the dispersion area is strongly dependent on both environmental and launch uncertainties, with their relative importance varying by flight altitude. For high-altitude launches, wind conditions at altitude were the dominant contributor to dispersion, followed by launch rail inclination and heading, whereas for low-altitude launches, launch uncertainties were the primary influence, with wind playing a secondary role. Other sources of uncertainty had a negligible effect on the dispersion area. These results highlight that dispersion uncertainty is regime-dependent and that mitigation efforts should prioritise reducing the most sensitive factors where possible. Improving launch alignment and reducing mechanical setup errors can significantly reduce low-altitude dispersion, while high-altitude predictions are inherently limited by wind uncertainty.

The analysis showed that the highest-energy landing events were associated with ballistic-type failures. These failures were typically located upstream of the launch site and frequently occurred outside the 98.9% confidence ellipsoids, posing a significant hazard to people and property. This highlights the importance of accounting for low-probability but high-consequence failure events in launch safety assessments, particularly for larger rockets, where impact energies are substantially higher. Consequently, efforts should focus on both reducing the probability of failures and mitigating their impact through exclusion zones or design changes.

A proof-of-concept method for reducing environmental uncertainty was demonstrated by estimating wind speed and heading from rocket flight data and feeding these estimates back into the dispersion analysis. The resulting predictions showed that the rocket landed within the predicted landing area, indicating that flight-derived atmospheric data can enhance dispersion estimation accuracy and may be applicable to successive flights.

This work enhances the understanding of how input uncertainty, including failures, affects rocket dispersion and provides a foundation for future research:

- Development of a probabilistic casualty model to quantitatively evaluate landing impact in amateur rocketry.
- Development of evidence-based casualty, impact, and damage threshold standards for amateur rocket safety assessments.
- Extended validation of Monte Carlo rocket trajectory simulation incorporating failure event modelling across multiple flight scenarios.

This work contributes to the development of evidence-based safety cases for rocket teams and launch site operations. By improving understanding of dispersion behaviour and uncertainty, the methods developed can support more robust risk assessments and operational decision-making within rocketry. In the UK, where launch opportunities are constrained by limited site dimensions and restricted altitude permissions, improved confidence in landing dispersion could enable more launches.

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A Additional Methodology

A.1 Drag Coefficient Estimation

The drag coefficients relative to the rocket's Mach number throughout flight, both with and without the motor producing thrust, are required for accurate simulations.

These can be measured experimentally or can be determined using computational methods. This section explains how to determine them experimentally using RAS Aero II [19], a combined aerodynamic and flight simulation program that can accurately predict subsonic-to-hypersonic aerodynamic coefficients for rockets and has been calibrated against NACA and NASA wind tunnel model data. RAS Aero II is also widely used in the amateur and student rocketry community and can be used to extract aerodynamic coefficients for use in other tools.

Step 1: OpenRocket Simulation

Create an accurate representation of the rocket's geometry in OpenRocket. This can be done directly in RAS Aero II, skipping steps 1 and 2, but it is generally easier to do in OpenRocket.

Step 2: Export to RAS Aero II

Export the OpenRocket design to RASAeroII from within OpenRocket ("File" > "Export As" > "RAS Aero II") as seen in fig. A.1. If prompted, you may need to remove or amend some unsupported features from OpenRocket before this is successful. This won't affect the drag coefficients unless you change external geometry.

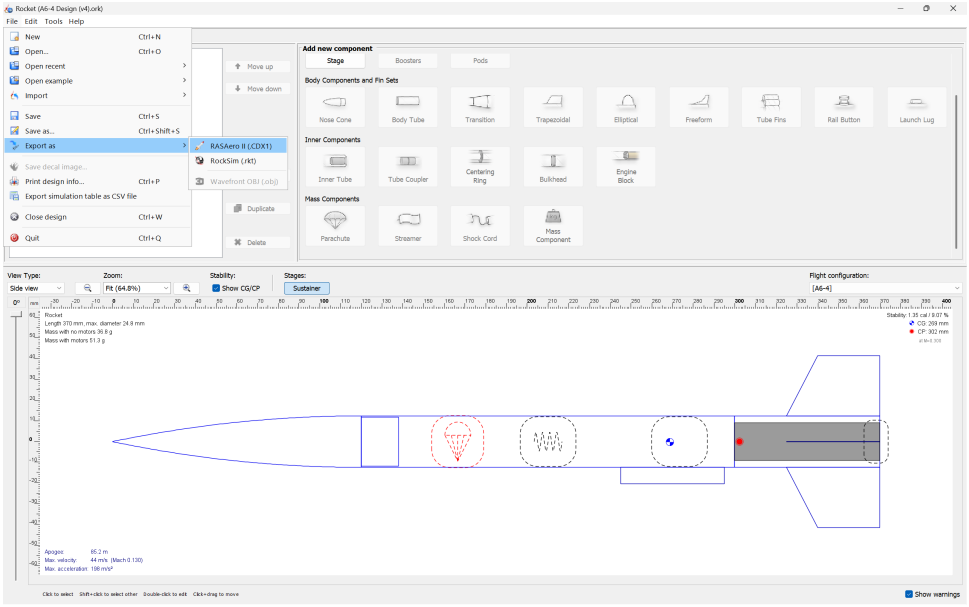


Figure A.1: Exporting to RAS Aero II from OpenRocket

Step 3: Open exported RAS Aero II file

Open RAS Aero II and then open the exported RAS Aero II file ("File" > "Open"). Once opened, confirm that the dimensions and geometry match those of the OpenRocket model.

Step 4: Add motor

Check that the correct motor has been selected (no motor may be selected) as shown in the blue circle of fig. A.2.

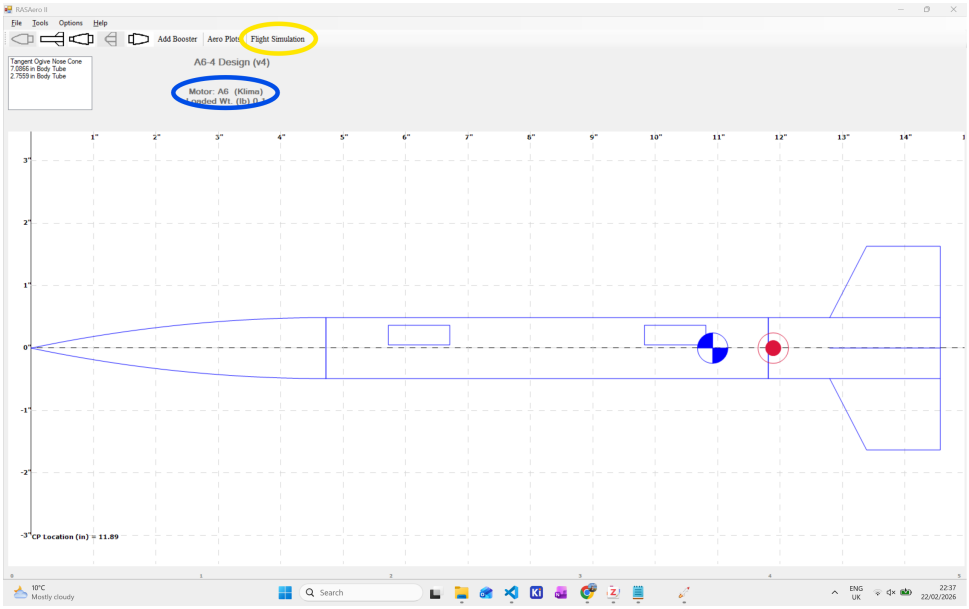


Figure A.2: Checking motor selection in RAS Aero II. Blue: Checking motor selection. Yellow: Opening flight simulations window

If the incorrect or no motor is selected, you should add the correct motor's thrust curve (.eng RASP file), which is available from thrustcurve.org [31] for most commercial motors, into RAS Aero II by selecting "File" > "Select Motor File". You then need to load it into the rocket by going to the "Flight Simulation" tab as seen in the yellow circle of fig. A.2 and then add this motor by selecting the motor slot and then the motor from the dropdown as shown in fig. A.3 in blue and yellow circles respectively. The flight simulation window can then be closed.

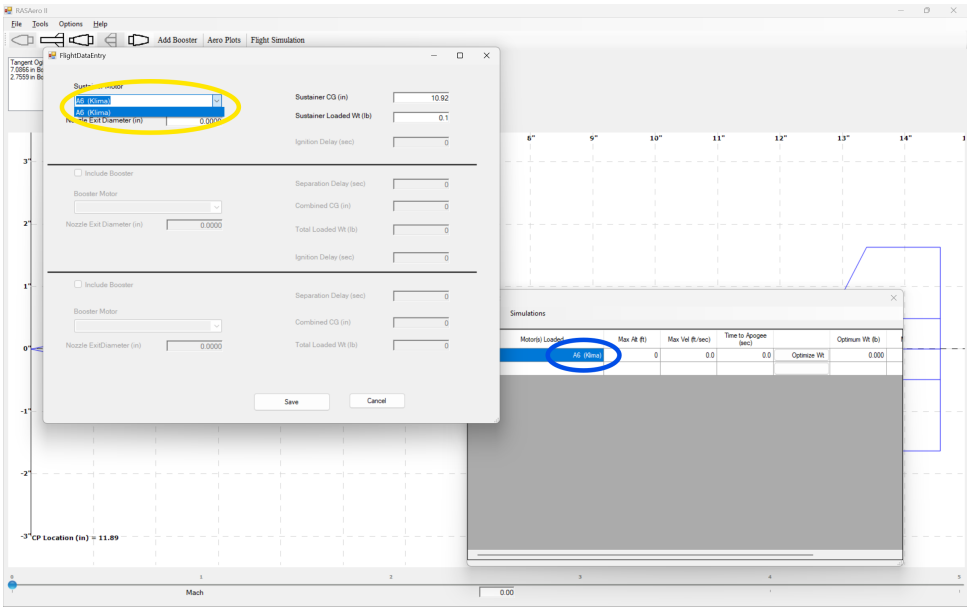


Figure A.3: Selecting a motor to use in RAS Aero II. Blue: Select motor slot. Yellow: Select which motor to use from the dropdown

Step 5: Create Aero Plots

Open the aerodynamic plots window by selecting the "Aero Plots" button seen in a blue circle in fig. A.4 and then export this data to a CSV file as seen ("File" > "Export" > "To CSV File").

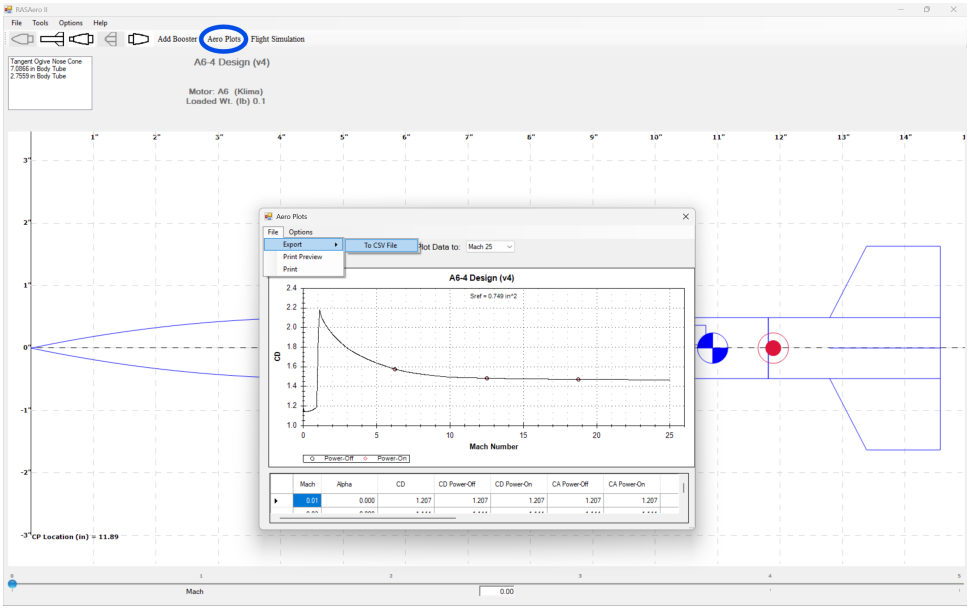


Figure A.4: Exporting aerodynamic plots from RAS Aero II. Blue: Opening the aero plots window.

Step 6: Extract useful data from csv

Use the extract_drag_curves.py script to extract 2 files (`powerOnDragCurve.csv` and `powerOffDragCurve.csv`) from the RAS Aero II exported file in the same directory.

```
$ python .\extract_drag_curves.py --input .\data\rockets\lpr\RASAeroOut.CSV
Extracted 'Mach' and 'CD Power-On' to C:\Users\rufus\OneDrive - The University of Manchester
    \Documents\University\Year 3\Individual Project\code\models\data\rockets\lpr\
    powerOnDragCurve.csv
Extracted 'Mach' and 'CD Power-Off' to C:\Users\rufus\OneDrive - The University of
    Manchester\Documents\University\Year 3\Individual Project\code\models\data\rockets\lpr\
    powerOffDragCurve.csv
```

A.2 Inertia Approximation

The moments of inertia of a rocket are required for accurate simulations, as they heavily influence its stability and dynamics. These are usually measured experimentally or estimated from a detailed CAD model of the vehicle. However, these methods are often not possible or take too much time, such as during the early design of a rocket.

This section explains how to estimate the approximate moment or inertias of a rocket based upon its mass and geometry using OpenRocket. This simple method assumes the centre of inertia is equal in the pitch and yaw axes and ignores the moment of inertia of the motors, as this is calculated separately within the model. It also assumes that an OpenRocket model of your rocket already exists and that you are familiar with OpenRocket.

Step 1: Add empty motor to OpenRocket

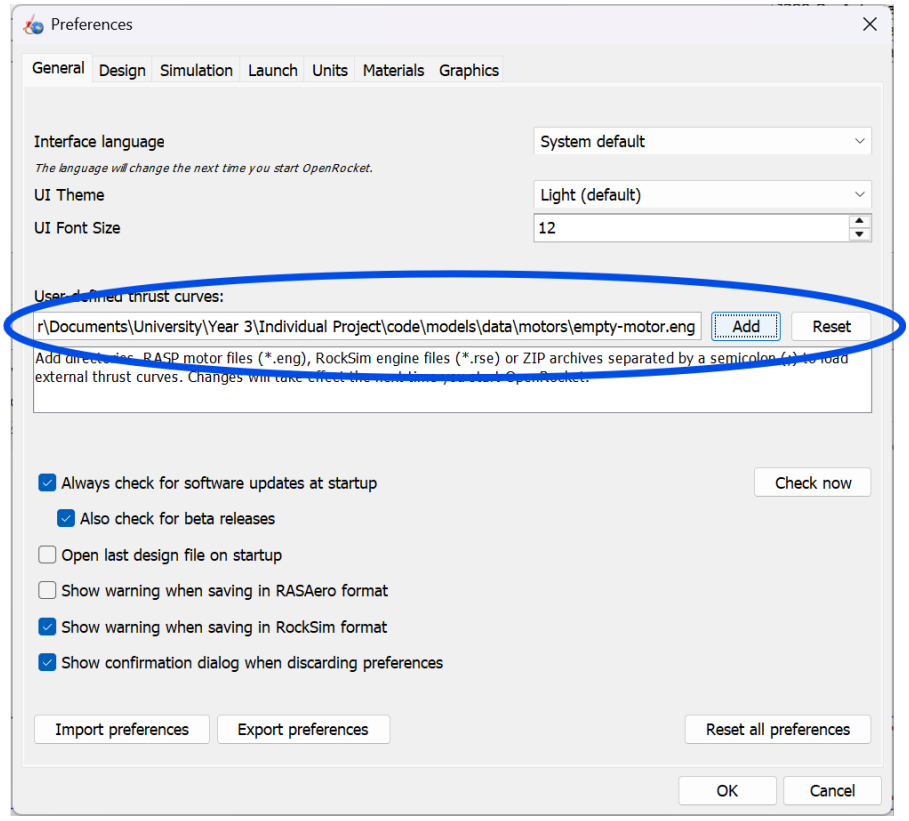


Figure A.5: Adding a motor to OpenRocket

Download the empty motor file from empty-motor.eng or create a mass-less (or near mass-less) motor yourself.

Open your OpenRocket model of the rocket. Navigate to user preferences ("Edit" > "Preferences") and then add the empty motors .eng file to OpenRocket as seen in fig. A.5 and then exit this menu.

Step 2: Add empty motor simulation

Create a new motor configuration with the new motor in the "Motors & Configuration" tab.

Step 3: Simulate and export

Simulate the flight with the new motor configuration and export the results to a CSV file with the "Plot / Export" button. Using the "Export Data" tab, make sure to include "Longitudinal moment of inertia" and "Rotational moment of inertia" as variables to export and select "Include flight events" as seen in fig. A.6. Press "Export" to save the data as a CSV file.

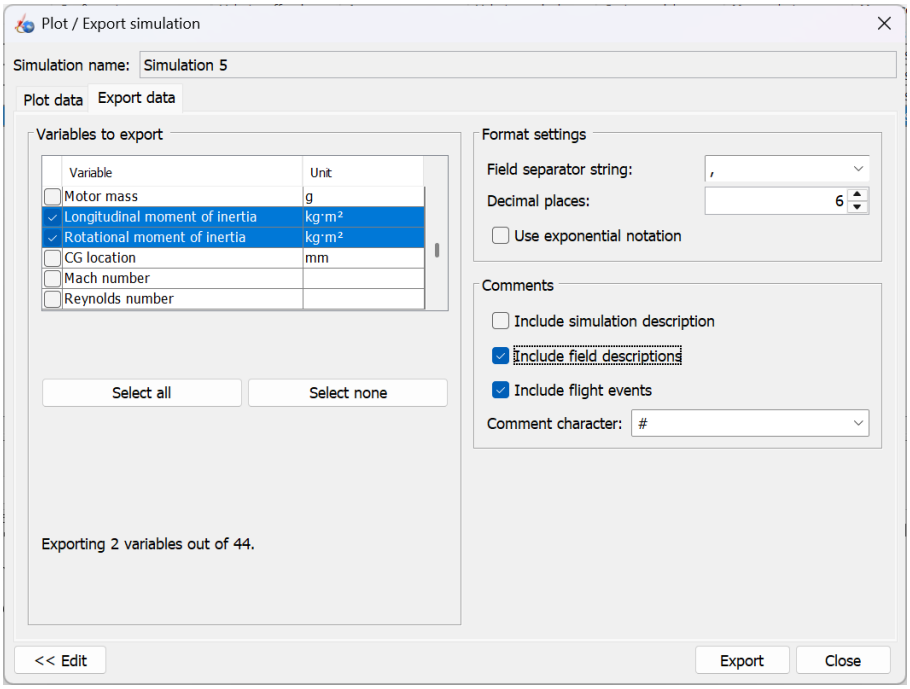


Figure A.6: Exporting flight data from OpenRocket

Step 4: Extract inertia values from simulations

Using the CSV files exported from OpenRocket, find the first row after motor burnout (or launch if there is no burnout event) and note the values of moment of inertia using the below relations about the dry centre of mass:

$$\text{Pitch Axis} = \text{Yaw Axis} = \text{Rotational Moment of Inertia} = I_{11} = I_{22}$$

$$\text{Roll Axis} = \text{Longitudinal Moment of Inertia} = I_{33}$$